

Physics-Informed Autoencoders for Parasitic Extraction in III-V Semiconductor Diodes

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Abstract

Accurate extraction of intrinsic and extrinsic parasitic parameters in high-frequency III-V semiconductor diodes is a critical bottleneck in mm-wave circuit design. While deep learning offers a pathway to accelerate this process, naive applications of neural networks to radio frequency (RF) data often fail due to physical misalignments, mathematical redundancies, and the curse of dimensionality. This paper presents a comprehensive, physics-first machine learning framework for the automated parameter extraction of GaN-based nanodiode Small Signal Equivalent Circuits (SSEC). The circuit is formulated as a 2-port network, retaining four extrinsic elements (C_{pad} , C_{ct} , L_{pad} , R_L) at their reference values while extracting the six variable intrinsic and semi-intrinsic parameters: $\{R_b, C_b, R_d, L_d, C_d, R_a\}$.

A multi-stage data pipeline is established. The 2-port S-parameter dataset (S_{11} , S_{21} , S_{22}) is first subjected to an empirical correlation analysis, post which S_{21} is dropped as the more fixture-contaminated parameter, while S_{11} and S_{22} are retained and concatenated in real-imaginary form to produce a 1,740-dimensional raw input vector. A Joint Complex PCA is then applied, compressing this vector to just 6 principal components that collectively capture $\approx 100\%$ of the total variance. Critically, this analysis reveals that L_d contributes a maximum absolute correlation of $|r| = 0.10$ with any principal component, establishing that L_d lies in the effective null space of the forward measurement operator at the target frequency range: its reactive impedance ωL_d is negligible relative to the parallel junction resistance R_d across the entire 40 MHz–43.5 GHz sweep.

For the extraction mechanism, two complementary neural architectures are proposed. The primary proposal (*Architecture Idea 1*) is a Physics-Informed Deterministic Autoencoder with a *triple-loss function*: a reconstruction loss operating in the 6-dimensional PCA space, a bottleneck physics-supervision loss, and a Kramers–Kronig (KK) consistency loss [11] that enforces electromagnetic causality on the physically reconstructed spectrum via a fixed inverse-PCA projection. The secondary proposal (*Architecture Idea 2*) is a Physics-Supervised Conditional Variational Autoencoder [4], which yields a full posterior distribution $p(\boldsymbol{\theta} \mid \mathbf{X}_{pca})$ over the six physical parameters—providing, among other diagnostics, a direct quantitative measure of the L_d unobservability through an inflated posterior variance on that parameter. A mechanistic interpretability framework—latent manifold traversal on a Smith Chart with a Poincaré disk geodesic metric, and frequency attribution mapping benchmarked against analytical sensitivities—completes the framework.

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1 Network Formulation: 2-Port vs. 1-Port

The foundation of any physics-informed machine learning model lies in the correct formulation of the underlying data. This project employs a 2-port network configuration within the Keysight Advanced Design System (ADS), yielding the full 2×2 scattering matrix $\{S_{11}, S_{21}, S_{12}, S_{22}\}$ (with $S_{12} = S_{21}$ by reciprocity). While a 1-port formulation is theoretically purer for a 2-terminal device, the 2-port topology was adopted due to its alignment with the reference circuit model [1] and practical simulation constraints. However, feeding all available S-parameters naively into a neural network introduces significant mathematical and physical misalignments that must be systematically corrected before machine learning can be applied.

1.1 The 2-Port Redundancy Problem and Empirical S-Parameter Selection

In a standard 2-port measurement, the scattering matrix encodes how a signal propagates *through* a device. For a 2-terminal device placed in a 2-port fixture, this generates three nominally independent observables: S_{11} (input reflection), S_{21} (forward transmission), and S_{22} (output reflection), with $S_{12} = S_{21}$. The theoretical arguments for redundancy among these are well established:

- **Fixture Pollution of S_{22} :** The output reflection coefficient S_{22} is dominated by the interaction between the device and the artificial $50\ \Omega$ output termination (TermG2). It predominantly encodes the fixture’s impedance environment rather than the intrinsic device parasitics, making it the least informative channel for parameter extraction.
- **Coupling Between S_{11} and S_{21} :** For a 2-terminal device, the forward transmission S_{21} is algebraically coupled to S_{11} through the device’s input impedance. Their signatures are therefore partially redundant, though not identical, since S_{21} carries additional through-path information absent in the input reflection alone.
- **Conflicting Optimization Gradients:** Training a neural network simultaneously on S_{11} , S_{21} , and S_{22} introduces conflicting gradient signals during backpropagation, since the three channels are not independent and their relative scales are not guaranteed to be compatible.

Rather than resolving S-parameter selection on purely theoretical grounds, an empirical **pairwise correlation analysis** was performed on the simulated dataset. For each pair of S-parameters, the Pearson correlation coefficient was computed independently for the magnitude, the real part, and the imaginary part of the spectra, yielding the following results:

Pair	$ r $ Magnitude	$ r $ Real part	$ r $ Imaginary part
S_{11} vs S_{21}	0.95	0.91	0.58
S_{11} vs S_{22}	0.92	0.91	0.58
S_{21} vs S_{22}	0.9949	1.000	1.000

The S_{21} – S_{22} correlation is the decisive result: real and imaginary part correlations of exactly 1.000 (to measurement precision) confirm that S_{21} and S_{22} are *linearly identical* in this dataset. This means there exist constants a, b, c, d such that $\text{Re}(S_{22}) = a \cdot \text{Re}(S_{21}) + b$ and $\text{Im}(S_{22}) = c \cdot \text{Im}(S_{21}) + d$ to numerical precision across all 20,000 simulations. Including both in the feature matrix would provide zero additional information while doubling the dimensionality of that channel pair.

The asymmetry in S_{11} – S_{21} imaginary part correlation ($|r| = 0.58$) compared to the real part ($|r| = 0.91$) is itself a physically meaningful observation: it is the signature of **non-circularity** of the complex S-parameter signals, meaning the real and imaginary parts do not share the same statistical structure. This will be accounted for in the PCA formulation of Section 5.1.

Decision: S_{21} is dropped from the feature set. The two retained channels are S_{11} and S_{22} . S_{22} is preferred over S_{21} here because S_{22} directly encodes the output reflection behavior of the device, which carries complementary spectral information to S_{11} , whereas S_{21} (forward transmission) is algebraically more redundant with S_{11} for a 2-terminal device in a shunt fixture topology. The final input representation is constructed from $\{\text{Re}(S_{11}), \text{Im}(S_{11}), \text{Re}(S_{22}), \text{Im}(S_{22})\}$.

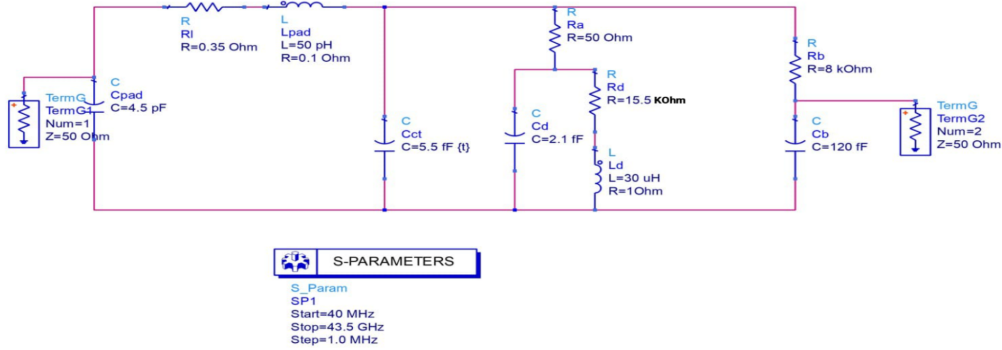


Figure 1: Initial 2-port schematic containing unnecessary output terminations, which pollutes the parameter space with fixture-dependent interactions.

1.2 Raw Input Representation and the Phase Wrap Problem

With S_{11} and S_{22} selected as the two retained channels, each sampled at $N_f = 435$ frequency points (40 MHz to 43.5 GHz, decimated to 100 MHz steps),

the raw S-parameter data per simulation sample is a pair of complex vectors $S_{11}(\omega), S_{22}(\omega) \in \mathbb{C}^{435}$. Before any further processing, these must be converted to a real-valued representation suitable for standard machine learning pipelines.

The natural complex representation offers two choices: polar form (magnitude and phase) or Cartesian form (real and imaginary parts). Polar form is **strongly contraindicated** due to the phase wrap problem. Phase is cyclical modulo 2π ; a physically smooth and continuous transition from -179° to $+179^\circ$ represents a mere 2° physical shift, but numerically appears as a violent 358° discontinuity. This artificial discontinuity corrupts the covariance structure of the data matrix, which is the foundational input to PCA, and would inject spurious high-variance directions into the principal components. It would equally disrupt gradient computation if used directly as neural network input.

The representation adopted here is therefore the **Cartesian (Real/Imaginary) form**, which oscillates smoothly and continuously between -1 and $+1$ with no discontinuities. The four real sub-vectors are concatenated in the following order to form the raw joint input per sample:

$$\begin{aligned} \mathbf{X}_{\text{raw}} = & \left[\underbrace{\text{Re}(S_{11}(f_1)), \dots, \text{Re}(S_{11}(f_{435}))}_{435 \text{ values}}, \right. \\ & \underbrace{\text{Im}(S_{11}(f_1)), \dots, \text{Im}(S_{11}(f_{435}))}_{435 \text{ values}}, \\ & \underbrace{\text{Re}(S_{22}(f_1)), \dots, \text{Re}(S_{22}(f_{435}))}_{435 \text{ values}}, \\ & \left. \underbrace{\text{Im}(S_{22}(f_1)), \dots, \text{Im}(S_{22}(f_{435}))}_{435 \text{ values}} \right] \in \mathbb{R}^{1740} \end{aligned} \quad (1)$$

This ordering keeps each channel’s real and imaginary parts adjacent, which improves the numerical conditioning of the sample covariance matrix used in the subsequent PCA step. Note that the phase wrap concern articulated above is a *data preparation* concern, not an architectural one: its resolution happens at the point of Equation (1), before any neural network processing occurs.

2 Related Work

This section situates the proposed methodology within the broader literature, identifying the conceptual threads from which this work draws and clearly delineating how it advances or diverges from each.

2.1 Physics-Informed Neural Networks and Supervised Latent Representations

The foundational idea of embedding physical laws directly into neural network training was formalized by Raissi, Perdikaris, and Karniadakis in their landmark work on Physics-Informed Neural Networks (PINNs) [3]. In that framework, the residuals of governing partial differential equations (PDEs)—evaluated at collocation points in the domain—are appended as penalty terms to the training

loss. This forces the network’s predicted field to simultaneously satisfy both the observed data and the governing differential constraints. PINNs have since demonstrated remarkable success in solving both forward and inverse problems in fluid mechanics, solid mechanics, and heat transfer.

The architecture proposed in this work occupies a distinct but philosophically related space. Rather than encoding differential operators as residual loss terms, physical knowledge is embedded through two complementary mechanisms: (1) explicit supervision of the encoder’s bottleneck to represent defined physical circuit parameters, which can be interpreted as a form of constrained latent space regularization; and (2) a Kramers–Kronig consistency penalty on the decoder’s reconstructed output, which constitutes a causality constraint entirely analogous in spirit to a PDE residual. The critical architectural distinction is that PINNs are designed primarily for forward problems—predicting a field given boundary conditions and a known PDE—whereas the core task here is an *inverse problem*: recovering the unknown parameter vector θ from spectral observations. The dual-loss autoencoder chain enforces physics on both the inverse map (encoder) and the forward map (decoder) simultaneously, a formulation that has no direct analogue in standard PINN literature.

2.2 Neural Network-Based Microwave and RF Modeling

The application of neural networks to microwave device modeling has a substantial history within the RF engineering community. Devabhaktuni, Yagoub, and Zhang [10] developed robust algorithms for the automated synthesis of neural network models of microwave components directly from S-parameter or bias-dependent measurement data, establishing knowledge-based neural network (KBNN) approaches as a credible alternative to physics-based simulation in the RF domain. These methods train the neural network as a black-box forward surrogate, mapping component geometries or bias conditions to scattering parameters.

The architecture proposed here inverts this classical paradigm. The decoder serves as the equivalent of a Devabhaktuni-style forward surrogate, mapping the six-dimensional circuit parameter vector θ back to the 6-dimensional PCA representation of the joint $\{S_{11}, S_{22}\}$ spectral space. Simultaneously, the encoder solves the inverse problem, recovering θ from an observed PCA-compressed spectrum. More critically, the dual-loss and triple-loss formulations ensure that the internal latent representation is interpretable directly in terms of actual circuit parasitics—a property entirely absent in classical black-box microwave neural network models, which make no guarantees about the physical meaning of their intermediate activations.

2.3 Variational Inference for Physical Inverse Problems

The extraction of SSEC parameters from S-parameter observations is, as established in Section 5.3, a fundamentally ill-posed inverse problem in which parameter degeneracies can produce nearly identical forward responses. Bayesian approaches address ill-posedness rigorously by replacing a point estimate of the un-

knowns with a full posterior distribution. Kingma and Welling’s Variational Autoencoder (VAE) [4] provides an elegant and computationally tractable approximation to Bayesian posterior inference via the Evidence Lower BOund (ELBO). The encoder of a VAE is trained to approximate the true (intractable) posterior $p(\mathbf{z} \mid \mathbf{x})$ with a tractable variational distribution $q_\phi(\mathbf{z} \mid \mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}(\mathbf{x}), \text{diag}(\boldsymbol{\sigma}^2(\mathbf{x})))$, allowing the entire framework to be optimized by gradient descent via the reparameterization trick.

In the context of physical system identification, VAE-style architectures provide the natural framework for quantifying the uncertainty in extracted parameters that arises from inherent parameter degeneracies, finite measurement bandwidth, and potential noise contamination—none of which are captured by a deterministic regression model that outputs only a point estimate. The Architecture Idea 2 proposed in Section 5.5 adapts the VAE to this setting by conditioning the latent prior on the physical ground-truth labels, creating a physically anchored probabilistic model of the extraction problem.

2.4 Space-Filling Experimental Design and Surrogate Modeling

The statistical framework underlying the data generation strategy in this work is the Design and Analysis of Computer Experiments (DACE) [9], a discipline concerned with efficiently probing the input space of computationally expensive simulators. Latin Hypercube Sampling, introduced by McKay, Beckman, and Conover [8], is the canonical space-filling design in this context, guaranteeing uniform marginal coverage of each input dimension with N samples regardless of the number of dimensions.

The classical surrogate model in DACE is the Gaussian Process (GP) emulator, which provides closed-form posterior inference over the simulator output. For this problem, however, GP emulators face fundamental scalability limitations: (i) computational complexity scales as $\mathcal{O}(N^3)$ in the number of training points, making $N = 20,000$ practically infeasible without sparse approximations; (ii) the output is a 1,740-dimensional spectral vector (pre-PCA), and standard GP kernels assume scalar outputs; and (iii) stationary GP covariance functions are poorly suited to the sharply frequency-dependent resonance behavior characteristic of SSEC impedance spectra. The neural network surrogate framework proposed here avoids all three limitations while simultaneously solving the inverse problem.

2.5 Causality and Dispersion Relations in Electromagnetic Modeling

The Kramers–Kronig (KK) relations, rigorously derived from the causality principle by Toll [11], establish that the real and imaginary parts of any linear, causal, passive system’s complex frequency response are not independent quantities: given one, the other is fully determined via a Hilbert transform. In RF and microwave engineering, KK consistency checks are routinely applied as a *post-hoc passivity verification* tool to assess the physical plausibility of measured or modeled S-parameters.

The key departure of the present work is the transformation of the KK relations from a verification criterion into an *active generative loss term* during neural network training. By penalizing the decoder for reconstructing non-causal S_{11} spectra, the KK loss term enforces a hard physical prior at training time. This approach bears a conceptual resemblance to the use of physical conservation laws as loss terms in PINNs [3], but is formulated in the frequency domain and targets the passivity and causality of a reconstructed complex spectrum rather than the residual of a differential equation.

3 Circuit Simplification for High-Frequency RF Extraction

The baseline topology for this project was derived from a comprehensive Small Signal Equivalent Circuit (SSEC) proposed by Pérez-Martín et al. [1]. However, initial simulations of this 1-port model across a high-frequency band (40 MHz to 43.5 GHz) yielded anomalous results: a nearly flatlined S_{11} magnitude curve hovering between -0.04 dB and -0.13 dB. An S_{11} near 0 dB indicates that roughly 99% of the injected signal is being reflected back to the source across the entire frequency band.

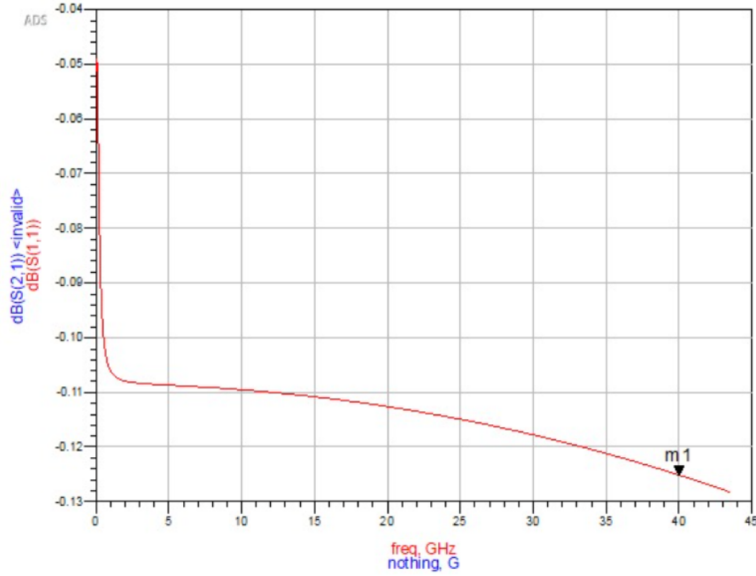


Figure 2: Initial S_{11} simulation displaying near-total reflection (flatline) on a rectangular magnitude plot.

3.1 Diagnosing the Flatline: Context of the Reference Literature

The error in the initial simulation stemmed from directly applying the *complete* equivalent circuit from the reference literature to a high-frequency RF simulation. The model in [1] was explicitly designed to capture trap-related

low-frequency dispersion. To achieve this, the authors introduced new elements to reproduce low-frequency impedance measurements (from 75 kHz to 30 MHz) and cryogenic behaviors.

Specifically, the reference model included an extra series R_b - C_b branch to model the influence of bulk traps, and a massive 30 μ H self-inductance (L_d) to model the current delay caused by the relaxation time of surface states.

While these components accurately reflect electron capture and release mechanisms at kilohertz frequencies, using them verbatim in a microwave/RF simulation introduces catastrophic errors. At 43.5 GHz, the impedance of the 30 μ H inductor becomes astronomically high:

$$Z_{L_d} = j2\pi f L_d = j2\pi(43.5 \times 10^9)(30 \times 10^{-6}) \approx j8.2 \text{ M}\Omega \quad (2)$$

In an RF context, this 8.2 M Ω impedance acts as a perfect RF choke. It completely blocks high-frequency AC current from entering the intrinsic branch, masking the true RF parasitics and causing the near-total signal reflection observed in Figure 2. These elements are mathematical pseudo-elements for low-frequency crystal defect modeling, not physical high-frequency parasitics.

3.2 Circuit Adaptation: Fixed Extrinsic Elements and Variable Intrinsic Parameters

As noted by Westlund et al. [2] and corroborated in [1], a simpler SSEC provides remarkable agreement with experimental data at high frequencies, with trap-related discrepancies only appearing below 1 GHz. The adaptation applied here proceeds in two stages.

Stage 1 — Excision of low-frequency trap pseudo-elements The three pseudo-elements introduced by [1] for low-frequency trap modeling are removed:

- **Removal of Surface Trap Inductance:** The 30 μ H inductor (L_d^{trap}) is deleted and replaced with a short circuit.
- **Removal of Bulk Trap Branch:** The parallel branch containing R_b^{trap} and C_b^{trap} is removed.
- **Removal of Output Termination:** The TermG2 port is removed from the intrinsic sub-circuit.

Stage 2 — Partitioning into fixed and variable elements Following guidance from the primary reference [1] and consistent with the approach adopted in related self-switching diode modeling work [2], the four extrinsic pad elements are held at their reference values throughout the dataset generation. They are not extraction targets:

C_{pad}	L_{pad}	R_L	C_{ct}
4.5 fF	50 pH	0.35 Ω	5.5 fF

The six remaining parameters are treated as the variable extraction targets, with their LHS sampling bounds set as follows:

Parameter	Physical meaning	LHS bounds
R_b	Bulk trap resistance	1 k Ω – 20 k Ω
C_b	Bulk trap capacitance	50 fF – 200 fF
R_d	Junction differential resistance	2 k Ω – 25 k Ω
L_d	Surface state inductance (RF remnant)	1 nH – 10 nH
C_d	Junction depletion capacitance	0.5 fF – 5.0 fF
R_a	Access resistance	10 Ω – 100 Ω

Note on L_d : The L_d retained here is distinct from the 30 μ H surface-trap inductance removed in Stage 1. It represents a residual surface-state reactive component in the RF-relevant nanohenry range. As will be established empirically in Section 5.1, this parameter lies in the effective null space of the 2-port S-parameter measurement at the target frequency range and is therefore unextractable from the available data.

3.3 Visualization: The Smith Chart for Complex Impedance

Beyond the physical circuit modifications, correct visualization is essential both for diagnosing simulation anomalies and for interpreting the trained model’s outputs. Plotting S_{11} on a standard rectangular dB magnitude chart discards all reactive information and squashes highly reflective data into a featureless flatline. The Smith Chart, which maps the complex reflection coefficient $\Gamma = S_{11}$ onto a 2D disk, reveals the full frequency-dependent trajectory of the device’s complex impedance. The constant-resistance and constant-reactance families of circles on the Smith Chart provide immediate physical intuition about whether a given feature is capacitive, inductive, or resistive in character. This representation forms the basis for the latent manifold traversal interpretability test described in Section 5.6.1.

4 Data Generation: Overcoming the Curse of Dimensionality

To train the physics-informed autoencoder, a comprehensive dataset mapping variations in the device’s physical parameters to corresponding high-frequency S_{11} curves must be generated. This requires simulating the 1-port equivalent circuit thousands of times across varying parasitic values.

4.1 The Grid Sweep Trap and Dimensionality

The variable parameter set contains $K = 6$ independent physical variables: $\{R_b, C_b, R_d, L_d, C_d, R_a\}$. A traditional nested parameter sweep (grid search) with $M = 20$ discrete steps per parameter results in:

$$N_{grid} = M^K = 20^6 = 6.4 \times 10^7 \text{ simulations} \quad (3)$$

Over 64 million simulations at even a fraction of a second each is categorically infeasible. Furthermore, a grid search on $K = 6$ parameters with M steps per dimension provides only M unique values per parameter regardless of how many total simulations are run—a catastrophically poor use of simulation budget. This motivates the LHS approach.

4.2 The Latin Hypercube Sampling (LHS) Solution

To efficiently map the multidimensional parameter space without triggering a combinatorial explosion, we employ **Latin Hypercube Sampling (LHS)** [8]. LHS is a stratified random sampling technique that provides mathematically guaranteed coverage of the parameter space, overcoming the critical flaws of both Grid Search and standard Monte Carlo methods.

4.2.1 Why LHS instead of Grid Search?

In a standard Grid Search, projecting M^K grid points onto any single parameter axis yields only M unique values. Even with 1.28×10^9 simulations, the network would only ever see 20 distinct values of any one parameter. LHS solves this: in a design of N samples, projection onto any single parameter axis yields exactly N distinct values uniformly distributed across the parameter’s range, maximizing information density per simulation [9].

4.2.2 Mathematical Formulation of LHS

Let $K = 6$ be the number of variable physical parameters and $N = 20,000$ be the total number of simulations. The sample value for the k -th parameter in the i -th simulation, denoted $x_k^{(i)}$, is generated as:

$$x_k^{(i)} = F_k^{-1} \left(\frac{\pi_k(i) - U_k^{(i)}}{N} \right) \quad (4)$$

where F_k^{-1} is the inverse CDF of the k -th parameter’s distribution (linear scaling between its physical bounds for a uniform prior), π_k is an independent random permutation of $\{1, \dots, N\}$ for dimension k , and $U_k^{(i)} \sim \mathcal{U}(0, 1)$ introduces within-stratum jitter to prevent sample alignment on a rigid grid.

Note: The number of variable parameters is $K = 6$: $\{R_b, C_b, R_d, L_d, C_d, R_a\}$.

The four fixed extrinsic elements (C_{pad} , L_{pad} , R_L , C_{ct}) are not swept and do not enter the LHS design.

4.2.3 Dataset Size Sufficiency

The target of $N = 20,000$ simulations is motivated by a coverage argument over the 6-dimensional parameter space. For a uniform LHS of N points in a K -dimensional unit hypercube, the fill distance scales approximately as:

$$h(N, K) \approx \left(\frac{K}{2\Gamma(\frac{K}{2} + 1)} \right)^{1/K} N^{-1/K} \quad (5)$$

For $K = 6$ and $N = 20,000$, this evaluates to a normalized fill distance of approximately 0.07—meaning no point in the normalized parameter space is further than roughly 7% of the total range from its nearest training sample. This is more than adequate for the smooth SSEC impedance manifold, and is consistent with the empirical finding (Section 5.1) that 6 principal components capture $\approx 100\%$ of variance, confirming no sharp discontinuities exist in the forward map over this parameter range.

Simultaneously, to address frequency array size, the frequency step size is decimated from 1 MHz to 100 MHz, reducing the vector length per channel from $\sim 43,460$ to $N_f = 435$ points. With two channels (S11 and S21) each contributing a real and imaginary part, the total raw input dimensionality per sample is $4 \times 435 = 1,740$, consistent with Equation (1).

4.3 Netlist Template Preparation in ADS

Before automated data generation can occur, a master simulation template must be extracted from the Keysight ADS environment. The Python automation script modifies and executes the raw text-based netlist directly.

The preparation protocol is as follows:

1. Build the 2-port RF schematic with the six variable elements $\{R_b, C_b, R_d, L_d, C_d, R_a\}$ and the four fixed elements ($C_{pad} = 4.5$ fF, $L_{pad} = 50$ pH, $R_L = 0.35 \Omega$, $C_{ct} = 5.5$ fF) in the ADS GUI.
2. Configure the S-Parameter simulation block: Start = 40 MHz, Stop = 43.5 GHz, Step = 100 MHz.
3. Execute the simulation once manually to generate the netlist.
4. Copy the `netlist.log` file and rename to `circuit_template.net`.
5. Replace the six variable component values with Python string formatting placeholders (e.g., `R:Rd _net1 0 R={Rd_val} Ohm`).
6. Ensure the S-parameter block contains `OutFile = "sim_output.s2p"` (2-port Touchstone format).

4.4 Automated Data Generation via Python

A Python script orchestrates the LHS data generation, injects sampled values into the netlist template, invokes the ADS simulation engine (hpeesofsim) headlessly, and aggregates results.

Listing 1: Python script for 6-parameter LHS generation and automated 2-port ADS simulation.

```
import os
import subprocess
import numpy as np
import pandas as pd
import skrf as rf
from scipy.stats import qmc

# --- 1. SETUP ---
NUM_SIMULATIONS = 20000
TEMPLATE_FILE = "circuit_template.net"
TEMP_NETLIST = "temp_run.net"
OUTPUT_S2P = "sim_output.s2p" # 2-port Touchstone
ADS_SIM_EXE = r"C:\Program Files\Keysight\ADS202X\bin\hpeesofsim.exe"

# Variable parameters: [Rb(Ohm), Cb(fF), Rd(Ohm), Ld(nH), Cd(fF), Ra(Ohm)]
lower_bounds = [1000, 50, 2000, 1.0, 0.5, 10]
upper_bounds = [20000, 200, 25000, 10.0, 5.0, 100]

# --- 2. LHS SAMPLING ---
sampler = qmc.LatinHypercube(d=len(lower_bounds))
sample = sampler.random(n=NUM_SIMULATIONS)
parameters = qmc.scale(sample, lower_bounds, upper_bounds)

with open(TEMPLATE_FILE, 'r') as f:
    template_text = f.read()

dataset = []

# --- 3. SIMULATION LOOP ---
for i in range(NUM_SIMULATIONS):
    rb, cb, rd, ld, cd, ra = parameters[i]

    current_netlist = template_text.format(
        Rb_val=rb, Cb_val=cb, Rd_val=rd,
        Ld_val=ld, Cd_val=cd, Ra_val=ra
    )
    with open(TEMP_NETLIST, 'w') as f:
        f.write(current_netlist)

    subprocess.run([ADS_SIM_EXE, TEMP_NETLIST],
                    stdout=subprocess.DEVNULL, stderr=subprocess.DEVNULL)

# --- 4. PARSE 2-PORT RESULTS ---
try:
    network = rf.Network(OUTPUT_S2P)
```

```

freqs    = network.f

# Retain S11 and S22; drop S21 (empirically confirmed
# redundant)
s11_re = network.s_re[:, 0, 0]
s11_im = network.s_im[:, 0, 0]
s22_re = network.s_re[:, 1, 1]
s22_im = network.s_im[:, 1, 1]

# Row: [6 labels] + [Re(S11) | Im(S11) | Re(S22) | Im(S22)]
row = [rb, cb, rd, ld, cd, ra]
row.extend(s11_re.tolist())
row.extend(s11_im.tolist())
row.extend(s22_re.tolist())
row.extend(s22_im.tolist())
dataset.append(row)

except Exception:
    pass

# --- 5. SAVE TO CSV ---
cols = ['Rb', 'Cb', 'Rd', 'Ld', 'Cd', 'Ra']
for f in freqs:
    cols.append(f'S11_Re_{f/1e9:.2f}GHz')
for f in freqs:
    cols.append(f'S11_Im_{f/1e9:.2f}GHz')
for f in freqs:
    cols.append(f'S22_Re_{f/1e9:.2f}GHz')
for f in freqs:
    cols.append(f'S22_Im_{f/1e9:.2f}GHz')

df = pd.DataFrame(dataset, columns=cols)
df.to_csv("ssec_training_data.csv", index=False)

```

Note: The code above has not been tested for compilation or runtime errors. It will require modification to match the exact netlist format of the local ADS installation. The total simulation time for 20,000 runs will be substantial.

The raw dataset is a flat matrix where the first 6 columns are physical target labels $\mathbf{Y}_{gt} \in \mathbb{R}^{N \times 6}$, and the subsequent 1,740 columns form the raw input matrix $\mathbf{X}_{raw} \in \mathbb{R}^{N \times 1740}$ as defined in Equation (1). Before this matrix is fed to the neural architecture, a dimensionality reduction step is applied, detailed in the following section.

Note: The code given above will require modifications based on the exact netlist format generated by the specific ADS version in use. It has not been tested for compilation or runtime errors and will take a significant amount of time to complete all 20,000 simulations.

5 Machine Learning Framework

The objective of this neural network is the continuous extraction of underlying physical parameter values from the compressed spectral representation produced by the Joint Complex PCA pipeline. The encoder portion of the architecture acts as the parameter extractor; the decoder acts as a physics surrogate, mapping extracted parameters back to the PCA-space spectral representation. A fixed, differentiable inverse-PCA projection then recovers the full 1,740-dimensional S-parameter spectrum from the decoder output for the purpose of applying physically motivated loss terms.

5.1 Joint Complex PCA: From 1740 Dimensions to 6

5.1.1 Theoretical Basis: Augmented Complex PCA for Non-Circular Signals

The raw input matrix $\mathbf{X}_{raw} \in \mathbb{R}^{N \times 1740}$ is a high-dimensional representation of the 6-parameter SSEC manifold. Feeding this directly to a neural encoder is feasible in principle but wasteful: the SSEC forward map $\mathcal{F}$ is smooth and the resulting data manifold is low-dimensional. A principled linear compression step is both computationally beneficial and physically diagnostic.

The two retained S-parameter channels, $S_{11}(\omega)$ and $S_{22}(\omega)$, are complex-valued random processes. Their joint representation as a 2-component complex vector $\mathbf{s}(\omega) = [S_{11}(\omega), S_{22}(\omega)]^T \in \mathbb{C}^2$ at each frequency is characterized by two covariance quantities:

$$\mathbf{C}_H = \mathbb{E}[\mathbf{s} \mathbf{s}^H] \quad (\text{Hermitian covariance}) \quad (6)$$

$$\mathbf{C}_P = \mathbb{E}[\mathbf{s} \mathbf{s}^T] \quad (\text{pseudo-covariance}) \quad (7)$$

A complex random vector is called **circular** (or proper) if $\mathbf{C}_P = \mathbf{0}$, meaning the real and imaginary parts are uncorrelated with equal variance. Standard complex PCA, which diagonalizes only $\mathbf{C}_H$, is optimal for circular signals. For **non-circular** (improper) signals, $\mathbf{C}_P \neq \mathbf{0}$, and the correct framework is **Widely Linear PCA**, which diagonalizes the augmented covariance:

$$\mathbf{C}_{aug} = \begin{bmatrix} \mathbf{C}_H & \mathbf{C}_P \\ \mathbf{C}_P^* & \mathbf{C}_H^* \end{bmatrix} \quad (8)$$

This augmented formulation processes both $\mathbf{s}$ and its conjugate $\mathbf{s}^*$, extracting principal components that fully exploit both the standard and pseudo-covariance structure.

The empirical evidence for non-circularity in the dataset is unambiguous: the inter-channel correlation between S_{11} and S_{22} is $|r|_{Re} = 0.92$ for the real part but only $|r|_{Im} = 0.58$ for the imaginary part. If the signals were circular, these two values would be equal. The asymmetry is a direct measurement of the pseudo-covariance $\mathbf{C}_P$.

Crucially, Widely Linear PCA on the complex vector $\mathbf{s}$ is provably equivalent to standard real PCA on the augmented real vector $[\text{Re}(\mathbf{s}), \text{Im}(\mathbf{s})]$ —which is

precisely the concatenation structure of Equation (1). The Joint Complex PCA performed on $\mathbf{X}_{raw}$ is therefore the theoretically correct method for non-circular complex signals, implemented via straightforward real-valued linear algebra.

5.1.2 Why Not Independent Per-Channel PCA?

The alternative approach—running four separate PCA instances on $\text{Re}(S_{11})$, $\text{Im}(S_{11})$, $\text{Re}(S_{22})$, $\text{Im}(S_{22})$ individually—diagonalizes four independent covariance matrices $\{\mathbf{C}_{kk}\}$ but explicitly discards the cross-covariance blocks $\{\mathbf{C}_{kl}, k \neq l\}$. Given that $|r|_{\text{Re}}(S_{11}, S_{22}) = 0.92$, these cross-covariance terms are far from negligible. Independent PCA would therefore discard exactly the inter-channel structure that most compactly encodes the shared physical parameters, producing principal components that are statistically suboptimal and physically misleading.

5.1.3 PCA Pipeline and Standardization

The full pipeline, applied once to the training set and then used as a fixed transform at inference time, proceeds as follows:

Step 1 — Standardization Each of the 1,740 input features is normalized to zero mean and unit variance using a `StandardScaler` fitted on the training set:

$$\tilde{X}_{raw,j} = \frac{X_{raw,j} - \mu_j}{\sigma_j}, \quad j = 1, \dots, 1740 \quad (9)$$

This is essential because different frequency bands and different S-parameter channels have different dynamic ranges. Without standardization, high-variance frequency bands would dominate the principal components purely due to scale rather than physical information content.

Step 2 — Joint Covariance and Eigendecomposition The sample covariance matrix $\hat{\mathbf{C}} \in \mathbb{R}^{1740 \times 1740}$ is computed from the standardized training matrix $\tilde{\mathbf{X}}_{raw}$. The eigendecomposition $\hat{\mathbf{C}} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$ (with eigenvalues sorted in descending order) gives the principal component directions $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_{1740}]$.

Step 3 — Projection to 6 components The PCA-compressed representation of sample i is:

$$\mathbf{x}_{pca}^{(i)} = \mathbf{V}_{:,1:6}^T \tilde{\mathbf{x}}_{raw}^{(i)} \in \mathbb{R}^6 \quad (10)$$

where $\mathbf{V}_{:,1:6}$ retains the 6 eigenvectors corresponding to the 6 largest eigenvalues.

Step 4 — Inverse PCA (used inside loss computation only) During training, the decoder reconstructs a 6-dimensional PCA-space vector $\hat{\mathbf{z}} \in \mathbb{R}^6$. To evaluate the Kramers–Kronig consistency loss (which requires access to the full spectrum split by channel), an inverse PCA projection recovers the approximate full 1,740-dimensional vector:

$$\hat{\mathbf{X}}_{1740} = \mathbf{V}_{:,1:6} \hat{\mathbf{z}} + \boldsymbol{\mu}_{scaler} \quad (11)$$

Since the 6 components capture $\approx 100\%$ of total variance, this reconstruction is effectively lossless for any sample drawn from the training distribution. The inverse PCA is a **fixed linear map** with no learnable parameters; it is therefore fully differentiable, and gradients of any loss computed after the inverse projection flow back through Equation (11) into the decoder weights without obstruction.

5.1.4 Empirical Results: Intrinsic Dimensionality and Manifold Structure

Running the joint complex PCA on the 20,000-sample dataset yields the following explained variance decomposition:

Component	Individual variance (%)	Cumulative variance (%)
PC1	~ 52	~ 52
PC2	~ 30	~ 82
PC3	~ 10	~ 92
PC4	~ 5	~ 97
PC5	~ 2	~ 99
PC6	~ 1	~ 100

Six components capture essentially 100% of the total variance. This is a profound result: the 1,740-dimensional raw input lies on a **6-dimensional linear subspace** of the ambient space, matching exactly the number of variable physical parameters. This confirms that the SSEC forward map, over the chosen frequency range and parameter bounds, produces a data manifold whose intrinsic dimensionality equals the number of free parameters—a signature of a well-posed, non-degenerate forward model (with important exceptions discussed below).

The PC–parameter correlation analysis further reveals which principal component carries information about which physical parameter. Computing the Pearson correlation $|r(PC_i, \theta_k)|$ between each of the 6 PCA scores and each of the 6 physical parameters yields a near-diagonal structure:

PC	Dominant parameter	$ r $ value
PC1	R_d	0.68
PC2	R_a	0.97
PC3	C_d	0.59
PC4	C_b	0.67
PC5	L_d	≤ 0.20
PC6	R_b	0.64

The near-diagonal structure confirms that the PCA basis has spontaneously organized itself approximately along the physical parameter axes. The clean assignment of PC2 to R_a ($|r| = 0.97$) is particularly striking and physically interpretable: R_a is a frequency-independent real-axis shift of Z_{in} , the simplest possible spectral signature, and the PCA correctly isolates it into a single dominant component.

5.1.5 The Unobservability of L_d : A Null-Space Finding

The most significant result of the PCA analysis is the complete absence of L_d from the correlation structure. The maximum absolute correlation between L_d and any of the 6 principal components is $|r| = 0.10$ —statistically indistinguishable from zero. This establishes, empirically, that L_d lies in the **effective null space of the forward measurement operator $\mathcal{F}$** at the target frequency range.

The physical explanation follows directly from the circuit topology. Within the intrinsic branch, L_d appears in series with R_d , giving a branch impedance:

$$Z_{branch}(\omega) = R_d + j\omega L_d \quad (12)$$

The relative magnitude of L_d 's contribution is characterized by:

$$\frac{|\omega L_d|}{R_d} = \frac{2\pi f \cdot L_d}{R_d} \quad (13)$$

Even at the top of the measurement band, $f = 43.5$ GHz, and taking typical values $L_d \sim 5$ nH and $R_d \sim 10$ k Ω :

$$\frac{2\pi \times 43.5 \times 10^9 \times 5 \times 10^{-9}}{10,000} = \frac{1366}{10,000} \approx 0.14 \quad (14)$$

The inductive reactance of L_d is at most $\sim 14\%$ of R_d at the highest frequency—and since R_d is simultaneously varying across the LHS range (2–25 k Ω), the L_d signature is completely swamped by the dominant R_d signature at every frequency point. No combination of frequency points in the 40 MHz–43.5 GHz sweep can isolate L_d 's contribution. The Jacobian column for L_d is effectively parallel to the R_d column across all frequencies, and the two are nearly linearly dependent.

This finding has direct architectural consequences:

- **Architecture Idea 1 (Deterministic):** L_d is fixed at its reference value and removed from the extraction target set. The effective target dimensionality reduces from 6 to 5: $\theta_{extracted} = \{R_b, C_b, R_d, C_d, R_a\}$. The encoder maps $\mathbb{R}^6 \rightarrow \mathbb{R}^5$ and the decoder maps $\mathbb{R}^5 \rightarrow \mathbb{R}^6$. Attempting to extract L_d from a deterministic encoder would produce an output uncorrelated with the true L_d value—essentially hallucinated predictions.

- **Architecture Idea 2 (CVAE):** L_d is retained as a target, but the CVAE’s posterior variance $\sigma_{L_d}^2(\mathbf{x}_{pca})$ is expected to be large and approximately constant regardless of the input, quantifying the fundamental unobservability. This constitutes a rigorous Bayesian statement of the null-space finding: the posterior over L_d remains close to the prior, indicating that the likelihood $p(\mathbf{x}_{pca} | \boldsymbol{\theta})$ is flat in the L_d direction. This capability—reporting uncertainty rather than hallucinating a value—is a key advantage of the probabilistic architecture.

5.2 Dataset Dimensionality and Structure After PCA

Following the joint complex PCA pipeline, the dataset is partitioned into the following tensors for neural network training:

- **PCA-compressed input ($\mathbf{X}_{pca}$):** A tensor of shape (20,000, 6). This is the primary input to the encoder. Each row is a 6-dimensional PCA score vector as defined in Equation (10).
- **Raw scaled input ($\tilde{\mathbf{X}}_{raw}$):** A tensor of shape (20,000, 1,740). Retained as a by-product of the PCA fit. Used in the no-PCA baseline (Condition 1 of the ablation study) and as the target space for the inverse-PCA KK loss computation.
- **Ground truth targets ($\mathbf{Y}_{gt}$):** A tensor of shape (20,000, 5) for Architecture Idea 1 (L_d fixed), or (20,000, 6) for Architecture Idea 2 (CVAE with L_d as uncertainty target).

5.3 The Inverse Extraction Problem and Its Ill-Posedness

Before specifying the architecture, it is essential to understand the mathematical nature of the problem being solved. The forward map of the ADS simulation, expressed in the PCA-compressed domain, is:

$$\mathcal{F} : \boldsymbol{\theta} \in \mathbb{R}^5 \longrightarrow \mathbf{x}_{pca}(\boldsymbol{\theta}) \in \mathbb{R}^6 \quad (15)$$

where $\boldsymbol{\theta} = \{R_b, C_b, R_d, C_d, R_a\}$ (with L_d fixed) and $\mathbf{x}_{pca}$ is the 6-dimensional PCA projection of the joint $\{S_{11}, S_{22}\}$ spectrum. This is an *overdetermined* forward map (5 inputs, 6 outputs), consistent with the fact that the 5 observable parameters generate a 5-dimensional manifold embedded in the 6-dimensional PCA space. The inverse problem—recovering $\boldsymbol{\theta}$ from an observed $\mathbf{x}_{pca}$ —seeks to invert this map.

5.3.1 Jacobian Analysis and Parameter Sensitivity Bands

The local behavior of the forward map near a given operating point $\boldsymbol{\theta}_0$ is characterized by its Jacobian $\mathbf{J} \in \mathbb{R}^{6 \times 5}$ in PCA space (or equivalently $\mathbf{J}_{full} \in \mathbb{C}^{N_f \times 5}$ in the original spectral space), whose (k, m) entry in the spectral domain is:

$$J_{k,m} = \left. \frac{\partial S_{11}(\omega_k; \boldsymbol{\theta})}{\partial \theta_m} \right|_{\boldsymbol{\theta}=\boldsymbol{\theta}_0} \quad (16)$$

The partial derivatives exhibit characteristic frequency dependencies consistent with the circuit topology:

$$\frac{\partial Z_{in}}{\partial R_d} \sim \frac{1}{(1 + j\omega R_d C_d)^2} \quad (\text{dominant at low-to-mid GHz}) \quad (17)$$

$$\frac{\partial Z_{in}}{\partial C_d} \sim \frac{-R_d^2 j\omega}{(1 + j\omega R_d C_d)^2} \quad (\text{most sensitive mid-GHz range}) \quad (18)$$

$$\frac{\partial Z_{in}}{\partial R_a} \sim 1 \quad (\text{frequency-independent, real axis}) \quad (19)$$

$$\frac{\partial Z_{in}}{\partial R_b} \sim \frac{1}{(1 + j\omega R_b C_b)^2} \quad (\text{bulk trap: low-to-mid GHz, similar to } R_d) \quad (20)$$

$$\frac{\partial Z_{in}}{\partial C_b} \sim \frac{-R_b^2 j\omega}{(1 + j\omega R_b C_b)^2} \quad (\text{similar band to } C_d, \text{ potential degeneracy}) \quad (21)$$

The sensitivity band diagram (Figure 3) should be updated to reflect the six-parameter configuration. The near-diagonal PC–parameter correlation structure observed empirically (Section 5.1) is the data-driven confirmation that these sensitivity bands are sufficiently distinct to make the extraction problem tractable for five of the six parameters.

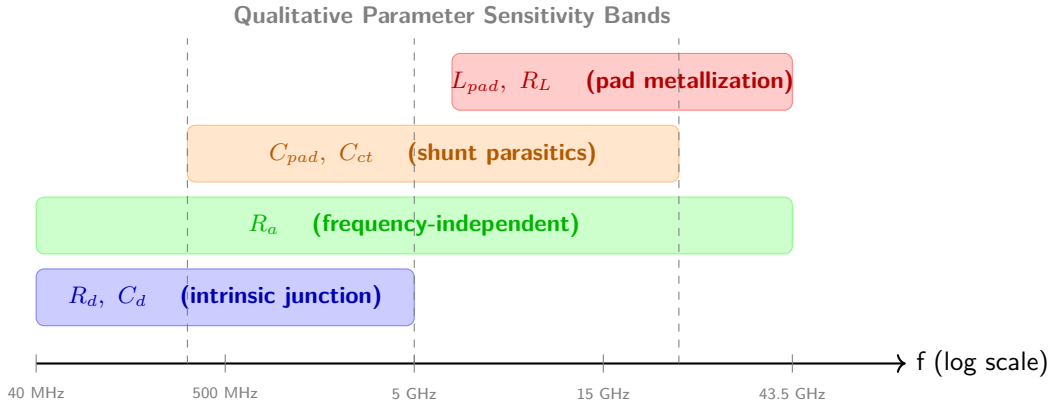


Figure 3: Qualitative frequency sensitivity bands for the seven SSEC parameters. Each parameter exerts its dominant influence over a distinct frequency sub-band, motivating the wide 40 MHz–43.5 GHz sweep. Overlap regions (e.g., C_{pad} and C_{ct}) represent potential ill-conditioning zones for the inverse extraction map.

5.3.2 Ill-Conditioning and Degenerate Parameter Pairs

The inverse problem is ill-posed (in the sense of Hadamard) when the Jacobian matrix $\mathbf{J}_{full}$ is rank-deficient or ill-conditioned. For the six-parameter circuit, the most susceptible degenerate pairs are (R_b, R_d) —both are resistances in parallel sub-branches with similar low-to-mid GHz sensitivity bands—and (C_b, C_d) —both are junction-type capacitances whose frequency signatures partially overlap. The PC–parameter correlation heatmap of Section 5.1 provides a direct empirical measurement of these degeneracies: PC5, which is the hardest-to-decode

component ($|r| \leq 0.20$ with any parameter), likely corresponds to the near-null direction formed by a linear combination of these degenerate pairs. The condition number $\kappa(\mathbf{J}_{full}^\dagger \mathbf{J}_{full})$ quantifies the global severity of this ill-conditioning.

5.3.3 The Dual-Loss as a Tikhonov Regularizer

The proposed physics-supervision loss on the bottleneck is not simply a convenience for making the latent space “interpretable.” It is, in precise mathematical terms, a **Tikhonov regularization** of the ill-posed inverse problem. Consider the standard Tikhonov-regularized objective:

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \|\mathcal{F}(\boldsymbol{\theta}) - \mathbf{S}_{11}^{obs}\|^2 + \lambda \|\boldsymbol{\theta} - \boldsymbol{\theta}_{prior}\|^2 \quad (22)$$

Setting $\boldsymbol{\theta}_{prior} = \mathbf{Y}_{gt}$ (the ground-truth label in the training set) and identifying the first term with $\mathcal{L}_{recon}$ and the second term with $\mathcal{L}_{phys}$, the dual-loss objective is exactly the empirical approximation to this regularized problem over the training distribution. The hyperparameter λ in the dual-loss formulation thus has a rigorous interpretation as the Tikhonov regularization strength. Understanding this connection also clarifies why the physics supervision *improves* the reconstruction loss during training: by stabilizing the gradient signal through the encoder, the regularizer prevents the optimizer from getting trapped in the degenerate parameter valleys that would otherwise correspond to near-singular Jacobian directions.

5.4 Architecture Idea 1: Physics-Informed Deterministic Autoencoder

The primary proposed architecture is a deterministic Physics-Informed Autoencoder (PIAE). The encoder $\mathcal{E}_\phi : \mathbb{R}^6 \rightarrow \mathbb{R}^5$ maps the 6-dimensional PCA input to the 5 extractable physical parameters $\{R_b, C_b, R_d, C_d, R_a\}$ (with L_d fixed). The decoder $\mathcal{D}_\psi : \mathbb{R}^5 \rightarrow \mathbb{R}^6$ maps those parameters back to the 6-dimensional PCA space, serving as a physics surrogate.

An important architectural note: unlike classical autoencoders where the encoder compresses a high-dimensional input to a low-dimensional bottleneck, the encoder here maps from a *lower-dimensional* PCA space (6D) to a *similarly-dimensioned* physical space (5D). It is not performing compression in the traditional sense—it is performing a learned **nonlinear change of basis** from the PCA coordinate system to the physical parameter coordinate system. The decoder then performs the inverse: a learned nonlinear map from physical parameter space back to PCA spectral space. The PCA-to-physics mapping is not linear because the forward map $\mathcal{F}$ is nonlinear, and thus the iso-manifolds of individual parameters in PCA space are curved surfaces, not hyperplanes.

5.4.1 Triple-Loss Supervision and the Kramers–Kronig Consistency Term

A standard unsupervised autoencoder trains solely on reconstruction: $\mathcal{L}_{recon} = \text{MSE}(\hat{\mathbf{z}}_{recon}, \mathbf{x}_{pca})$, operating in the 6-dimensional PCA space. This leaves the

bottleneck completely unconstrained with respect to physical meaning. The PIAE adds a physics-supervision term $\mathcal{L}_{phys}$, and further introduces a novel third term: the **Kramers–Kronig Consistency Loss** $\mathcal{L}_{KK}$.

For any linear, passive, causal network, the real and imaginary parts of each S-parameter are related by the Hilbert transform through the Kramers–Kronig relations [11]:

$$\text{Re}[S(\omega)] = \mathcal{H}\{\text{Im}[S]\}(\omega), \quad \text{Im}[S(\omega)] = -\mathcal{H}\{\text{Re}[S]\}(\omega) \quad (23)$$

These relations hold separately for S_{11} and S_{22} (each is an entry of a causal scattering matrix). A neural network decoder trained without a causality constraint can reconstruct PCA-space vectors that, when projected back to the spectral domain, are physically non-causal—even if they achieve low reconstruction loss in PCA space.

The Inverse PCA Bridge The KK relations operate on $\text{Re}(S(\omega))$ and $\text{Im}(S(\omega))$ as functions of frequency. The decoder output $\hat{\mathbf{z}} \in \mathbb{R}^6$, however, lives in PCA space—a linear mixture of all four sub-vectors across all frequencies simultaneously. PCA components individually have no frequency-domain interpretation. The KK loss therefore cannot be applied directly to $\hat{\mathbf{z}}$.

To bridge this, the fixed inverse PCA transform of Equation (11) is applied inside the loss computation to recover an approximate 1,740-dimensional spectrum:

$$\hat{\mathbf{X}}_{1740} = \mathbf{V}_{:,1:6} \hat{\mathbf{z}} + \boldsymbol{\mu}_{scaler} \in \mathbb{R}^{1740} \quad (24)$$

This vector is then split according to the concatenation structure of Equation (1) into four sub-vectors:

$$\hat{\mathbf{X}}_{1740} \longrightarrow (\hat{\mathbf{R}}_{11}, \hat{\mathbf{I}}_{11}, \hat{\mathbf{R}}_{22}, \hat{\mathbf{I}}_{22}) \in \mathbb{R}^{435} \times \mathbb{R}^{435} \times \mathbb{R}^{435} \times \mathbb{R}^{435} \quad (25)$$

where $\hat{\mathbf{R}}_{11} = \hat{\mathbf{X}}_{1740}[0 : 435]$, $\hat{\mathbf{I}}_{11} = \hat{\mathbf{X}}_{1740}[435 : 870]$, $\hat{\mathbf{R}}_{22} = \hat{\mathbf{X}}_{1740}[870 : 1305]$, and $\hat{\mathbf{I}}_{22} = \hat{\mathbf{X}}_{1740}[1305 : 1740]$.

The discrete Hilbert transform is implemented via the FFT:

$$\hat{\mathcal{H}}\{x\}[k] = \mathcal{F}^{-1}(-j \cdot \text{sgn}(k) \cdot \mathcal{F}\{x\}[k]) \quad (26)$$

The KK consistency loss is then computed separately for each retained S-parameter channel and summed:

$$\begin{aligned} \mathcal{L}_{KK} = & \underbrace{\frac{1}{N_f} \left(\|\hat{\mathbf{R}}_{11} - \hat{\mathcal{H}}\{\hat{\mathbf{I}}_{11}\}\|^2 + \|\hat{\mathbf{I}}_{11} + \hat{\mathcal{H}}\{\hat{\mathbf{R}}_{11}\}\|^2 \right)}_{\mathcal{L}_{KK,S_{11}}} \\ & + \underbrace{\frac{1}{N_f} \left(\|\hat{\mathbf{R}}_{22} - \hat{\mathcal{H}}\{\hat{\mathbf{I}}_{22}\}\|^2 + \|\hat{\mathbf{I}}_{22} + \hat{\mathcal{H}}\{\hat{\mathbf{R}}_{22}\}\|^2 \right)}_{\mathcal{L}_{KK,S_{22}}} \end{aligned} \quad (27)$$

Since the inverse PCA is a fixed linear map with no learnable parameters, it is fully differentiable. Gradients of $\mathcal{L}_{KK}$ flow back through the inverse PCA

into the decoder weights ψ without obstruction:

$$\frac{\partial \mathcal{L}_{KK}}{\partial \psi} = \frac{\partial \mathcal{L}_{KK}}{\partial \hat{\mathbf{X}}_{1740}} \cdot \underbrace{\frac{\partial \hat{\mathbf{X}}_{1740}}{\partial \hat{\mathbf{z}}}}_{\mathbf{V}_{:,1:6}} \cdot \frac{\partial \hat{\mathbf{z}}}{\partial \psi} \quad (28)$$

Note that $\mathcal{L}_{KK}$ is applied *exclusively to the decoder output*. It acts as a physical prior on the generative model, ensuring the decoder only produces spectra that satisfy electromagnetic causality. This is directly analogous to PDE residual terms in PINNs [3], but formulated in the frequency domain.

The complete triple-loss objective for Architecture Idea 1 is:

$$\mathcal{L}_{total} = \underbrace{\text{MSE}(\hat{\mathbf{z}}_{recon}, \mathbf{x}_{pca})}_{\mathcal{L}_{recon}} + \lambda_1 \underbrace{\text{MSE}(\hat{\mathbf{Y}}_{bott}, \mathbf{Y}_{gt})}_{\mathcal{L}_{phys}} + \lambda_2 \underbrace{\mathcal{L}_{KK}(\hat{\mathbf{z}}_{recon})}_{\text{causality via inv. PCA}} \quad (29)$$

where $\hat{\mathbf{Y}}_{bott} = \mathcal{E}_\phi(\mathbf{x}_{pca})$ is the encoder’s 5-dimensional bottleneck output (the extracted physical parameters), and $\hat{\mathbf{z}}_{recon} = \mathcal{D}_\psi(\hat{\mathbf{Y}}_{bott})$ is the decoder’s reconstructed 6-dimensional PCA vector.

Note: MSE is not mandatory as the primary cost function. If the parameter distributions exhibit heavy tails or the reconstruction loss landscape is poorly conditioned, alternatives such as Huber loss or a frequency-weighted MSE (with weights $w(f) \propto |\partial S_{11}/\partial \theta|$ concentrated on the most diagnostically informative frequency bands) should be explored.

5.4.2 The λ_1, λ_2 Selection Problem and Constrained Optimization Formulation

The scalars $\lambda_1 \geq 0$ and $\lambda_2 \geq 0$ are the most consequential hyperparameters in the triple-loss framework, and their selection deserves rigorous treatment. At a surface level, Equation (29) appears to be a standard weighted sum minimization — but with the constraint $\lambda_1, \lambda_2 \geq 0$, it admits a cleaner interpretation as a constrained optimization problem, which opens a distinct set of solution strategies.

Constrained Optimization View Observe that Equation (29) can be rewritten as a primary objective with two inequality-constrained subsidiary objectives. Define:

$$\min_{\phi, \psi} \mathcal{L}_{recon}(\phi, \psi) \quad \text{subject to} \quad \mathcal{L}_{phys}(\phi) \leq \epsilon_1, \quad \mathcal{L}_{KK}(\psi) \leq \epsilon_2 \quad (30)$$

By the method of Lagrange multipliers, the Lagrangian of Equation (30) is exactly:

$$\mathcal{L}_{Lagrangian} = \mathcal{L}_{recon} + \lambda_1(\mathcal{L}_{phys} - \epsilon_1) + \lambda_2(\mathcal{L}_{KK} - \epsilon_2) \quad (31)$$

At the KKT optimality conditions, $\lambda_1 \geq 0$ and $\lambda_2 \geq 0$ are the Lagrange multipliers enforcing the physical accuracy and causality constraints respectively.

The non-negativity requirement is not simply a modelling choice — it is a **necessary condition for primal feasibility**: a negative λ_i would penalize the objective for being *too physically accurate* or *too causal*, which is nonsensical. This formulation has two important consequences:

1. The constraint thresholds ϵ_1 and ϵ_2 (the acceptable levels of physical error and KK residual) are the true design parameters of the problem, not λ_1 and λ_2 directly. Given a target ϵ_1 on physical parameter error, there exists a corresponding optimal λ_1^* that enforces it; the task of hyperparameter search is fundamentally a search over (ϵ_1, ϵ_2) in the dual problem.
2. When one of the constraints becomes inactive — for instance, when $\mathcal{L}_{KK}$ is already negligible because the decoder has learned causal responses naturally — the corresponding KKT complementary slackness condition $\lambda_2 \cdot (\mathcal{L}_{KK} - \epsilon_2) = 0$ implies $\lambda_2 \rightarrow 0$. This is the precise condition under which the KK loss becomes redundant and can be dropped from training.

Practical Selection Strategies Three practical strategies are proposed in increasing order of computational cost and rigour:

Strategy 1 — Grid Search with Logarithmic Spacing. Given that $\lambda_1, \lambda_2 \geq 0$, a standard grid search over a logarithmic grid is the simplest approach:

$$\lambda_i \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^0, 10^1\} \quad (32)$$

For a 6×6 grid of (λ_1, λ_2) pairs, this yields 36 training runs. Each run is evaluated on a validation set using MAPE on the 5 extracted parameters as the primary metric. The grid is searched in two passes: a coarse pass over the full range, followed by a fine pass around the best-performing coarse configuration. This is computationally tractable given the small input dimensionality (6D) and architecture size.

Strategy 2 — Kendall Homoscedastic Uncertainty Weighting [5]. Each loss term is re-parameterized as a task with an associated learnable log-variance $\log \sigma_i^2$:

$$\mathcal{L}_{total} = \frac{\mathcal{L}_{recon}}{2\sigma_1^2} + \frac{\mathcal{L}_{phys}}{2\sigma_2^2} + \frac{\mathcal{L}_{KK}}{2\sigma_3^2} + \log \sigma_1 + \log \sigma_2 + \log \sigma_3 \quad (33)$$

The parameters $\{\sigma_i\}$ are learned jointly with the network weights ϕ, ψ . The log terms prevent the degenerate solution $\sigma_i \rightarrow \infty$. The effective weight on task i is $\lambda_i^{eff} = 1/(2\sigma_i^2)$, which is guaranteed non-negative by construction. This is the cleanest implementation of the non-negativity constraint, and it requires no hyperparameter tuning beyond initializing the $\{\sigma_i\}$ near 1.

Strategy 3 — GradNorm Adaptive Balancing [6]. GradNorm normalizes the per-task gradient magnitudes at the last shared layer to prevent any single task from dominating. Define $G_i(t) = \|\nabla_W[w_i \mathcal{L}_i(t)]\|_2$ and the relative inverse training rate $r_i(t) = \tilde{\mathcal{L}}_i(t)/\tilde{\mathcal{L}}(t)$, where $\tilde{\mathcal{L}}_i = \mathcal{L}_i(t)/\mathcal{L}_i(0)$. GradNorm updates the weights by minimizing:

$$\mathcal{L}_{GradNorm} = \sum_i |G_i(t) - \bar{G}(t) \cdot r_i(t)^\alpha|_1 \quad (34)$$

with respect to $\{w_i\}$ only (network weights held fixed during this sub-step). The non-negativity of w_i must be explicitly enforced (e.g., via a softplus parameterization $w_i = \log(1 + \exp(\tilde{w}_i))$) to respect the Lagrangian structure.

Recommendation: Start with Strategy 1 (grid search) for initial experiments to establish a reference. If training instability is observed near the λ_1 – λ_2 boundaries, switch to Strategy 2 (Kendall weighting), which is the most principled and requires the least tuning. Strategy 3 is recommended only if grid search and Kendall weighting both exhibit slow convergence due to task-gradient imbalance.

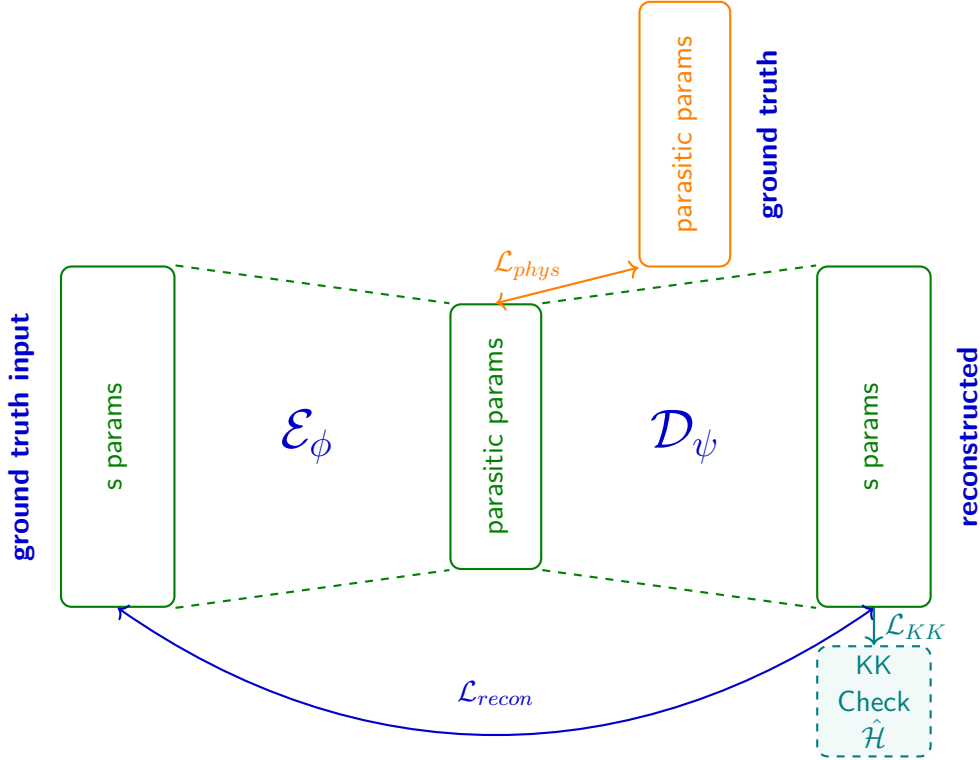


Figure 4: Triple-loss PIAE architecture (Idea 1). The encoder $\mathcal{E}_\phi : \mathbb{R}^6 \rightarrow \mathbb{R}^5$ maps the PCA-compressed input to the 5 extractable physical parameters. The decoder $\mathcal{D}_\psi : \mathbb{R}^5 \rightarrow \mathbb{R}^6$ reconstructs the PCA-space vector. $\mathcal{L}_{recon}$ enforces PCA-space reconstruction fidelity; $\mathcal{L}_{phys}$ supervises the 5-dimensional bottleneck (acting as Tikhonov regularization on the ill-posed inverse problem); $\mathcal{L}_{KK}$ enforces Kramers–Kronig causality on the decoder output after inverse-PCA projection to the full $\{S_{11}, S_{22}\}$ spectral domain.

5.4.3 Feature Normalization

Because the physical parameter values vary across vastly different magnitudes (e.g., $R_d \sim 10^3 \, \Omega$ while $C_d \sim 10^{-15} \, \text{F}$), passing unscaled targets into $\mathcal{L}_{phys}$ will

induce severe gradient explosion. Prior to training, both $\mathbf{X}$ and $\mathbf{Y}$ are independently scaled using Min-Max Normalization:

$$Y_{scaled,k} = \frac{Y_k - Y_{k,\min}}{Y_{k,\max} - Y_{k,\min}} \quad (35)$$

bounding all target values strictly within $[0, 1]$. During inference, the bottleneck output is passed through the inverse transform to recover true physical magnitudes.

5.4.4 Architecture Variants: Spectral Encoders

The input to the encoder, $\mathbf{X}_{input} \in \mathbb{R}^{870}$, is an ordered frequency-domain sequence. A fully-connected MLP encoder treats all 870 dimensions as exchangeable, ignoring the structural information that *adjacent frequencies are highly correlated* due to the smooth resonance behavior of SSEC impedances. Two architecturally motivated alternatives are worth evaluating:

Note on Input Dimensionality and Architecture Choice With the PCA compression reducing the input to 6 dimensions, the classical motivation for 1D CNN or FNO encoders (exploiting local spectral correlation across hundreds of frequency slots) is substantially reduced. The PCA components are no longer ordered frequency samples — they are abstract linear combinations of the entire spectrum. A standard MLP encoder is therefore the natural and recommended choice for the 6D PCA input. However, two alternatives remain relevant for the *no-PCA baseline* (Condition 1 of the ablation study, where the full 1,740-dimensional raw input is used directly):

1D Convolutional Encoder: Treats $\mathbf{X}_{raw}$ as a 4-channel signal ($\text{Re}(S_{11})$, $\text{Im}(S_{11})$, $\text{Re}(S_{22})$, $\text{Im}(S_{22})$) over $N_f = 435$ frequency positions. Convolutional layers exploit local spectral smoothness, reducing parameter count vs. a flat MLP while providing a physically motivated inductive bias.

Fourier Neural Operator (FNO) Encoder (Li et al. [7]): Operates natively in frequency space via learned truncated Fourier-mode transforms:

$$v_\ell(k) = \mathcal{F}^{-1}(\mathbf{R}_\ell \cdot \mathcal{F}\{u_\ell\}(k)) + \mathbf{W}_\ell u_\ell \quad (36)$$

where $\mathbf{R}_\ell \in \mathbb{C}^{d_{in} \times d_{out} \times k_{max}}$ is a learned complex weight tensor. Since $\mathbf{X}_{raw}$ is already a frequency-domain signal sampled on a uniform 100 MHz grid, FNO layers process it in its natural representation. This variant is most relevant for the no-PCA baseline.

5.4.5 Training Algorithm

Algorithm 1 Physics-Informed Triple-Loss Autoencoder Training (Idea 1)

Require: Dataset $\mathcal{D} = \{(\mathbf{x}_{pca}^{(i)}, \mathbf{Y}^{(i)})\}_{i=1}^N$; fixed matrices $\mathbf{V}_{:,1:6}$, $\boldsymbol{\mu}_{scaler}$ from PCA;
Epochs E ; Batch B ; LR α ; $\lambda_1, \lambda_2 \geq 0$

Ensure: Optimized encoder $\mathcal{E}_\phi : \mathbb{R}^6 \rightarrow \mathbb{R}^5$ and decoder $\mathcal{D}_\psi : \mathbb{R}^5 \rightarrow \mathbb{R}^6$

```

1: Initialize  $\phi, \psi$  using Xavier/Glorot initialization
2: for epoch  $e = 1$  to  $E$  do
3:   Shuffle  $\mathcal{D}$ 
4:   for each mini-batch  $\{\mathbf{x}_b, \mathbf{Y}_b\} \subset \mathcal{D}$  do
5:     // Forward pass
6:      $\hat{\mathbf{Y}}_{bott} \leftarrow \mathcal{E}_\phi(\mathbf{x}_b)$  // 5 extracted parameters
7:      $\hat{\mathbf{z}}_{recon} \leftarrow \mathcal{D}_\psi(\hat{\mathbf{Y}}_{bott})$  // 6D PCA reconstruction
8:     // Inverse PCA to recover full 1740D spectrum for KK check
9:      $\hat{\mathbf{X}}_{1740} \leftarrow \mathbf{V}_{:,1:6} \hat{\mathbf{z}}_{recon} + \boldsymbol{\mu}_{scaler}$ 
10:     $\hat{\mathbf{R}}_{11} \leftarrow \hat{\mathbf{X}}_{1740}[0:435]$ ;  $\hat{\mathbf{I}}_{11} \leftarrow \hat{\mathbf{X}}_{1740}[435:870]$ 
11:     $\hat{\mathbf{R}}_{22} \leftarrow \hat{\mathbf{X}}_{1740}[870:1305]$ ;  $\hat{\mathbf{I}}_{22} \leftarrow \hat{\mathbf{X}}_{1740}[1305:1740]$ 
12:    // Triple-loss computation
13:     $\mathcal{L}_{recon} \leftarrow \frac{1}{B} \sum_j \|\mathbf{x}_b^{(j)} - \hat{\mathbf{z}}_{recon}^{(j)}\|_2^2$ 
14:     $\mathcal{L}_{phys} \leftarrow \frac{1}{B} \sum_j \|\mathbf{Y}_b^{(j)} - \hat{\mathbf{Y}}_{bott}^{(j)}\|_2^2$ 
15:     $\mathcal{L}_{KK} \leftarrow \frac{1}{N_f} \left( \|\hat{\mathbf{R}}_{11} - \hat{\mathcal{H}}\{\hat{\mathbf{I}}_{11}\}\|^2 + \|\hat{\mathbf{I}}_{11} + \hat{\mathcal{H}}\{\hat{\mathbf{R}}_{11}\}\|^2 \right.$ 
16:       $\left. + \|\hat{\mathbf{R}}_{22} - \hat{\mathcal{H}}\{\hat{\mathbf{I}}_{22}\}\|^2 + \|\hat{\mathbf{I}}_{22} + \hat{\mathcal{H}}\{\hat{\mathbf{R}}_{22}\}\|^2 \right)$ 
17:     $\mathcal{L}_{total} \leftarrow \mathcal{L}_{recon} + \lambda_1 \mathcal{L}_{phys} + \lambda_2 \mathcal{L}_{KK}$ 
18:    Compute  $\nabla_\phi \mathcal{L}_{total}$ ,  $\nabla_\psi \mathcal{L}_{total}$ ; update  $\phi, \psi$  via gradient descent
19:   end for
20: end for
21: return  $\mathcal{E}_\phi, \mathcal{D}_\psi$ 

```

5.5 Architecture Idea 2: Physics-Supervised Conditional Variational Autoencoder

5.5.1 Motivation: The Posterior Over Physical Parameters

Architecture Idea 1 produces a single deterministic point estimate $\hat{\boldsymbol{\theta}} = \mathcal{E}_\phi(\mathbf{x}_{pca}) \in \mathbb{R}^5$. As established in Section 5.3 and confirmed empirically in Section 5.1, the inverse extraction map $\mathcal{F}^{-1}$ is ill-posed: the degenerate parameter pairs (R_b, R_d) and (C_b, C_d) produce partially overlapping spectral signatures, and a single observed PCA vector $\mathbf{x}_{pca}$ is consistent with a continuous *manifold* of parameter vectors $\boldsymbol{\theta}$. A deterministic encoder collapses this manifold to one point, discarding all information about which other parameter combinations are equally compatible with the observation.

Architecture Idea 2 addresses this by modeling the extraction problem as a Bayesian inference task. Instead of seeking a point estimate, the goal is the full posterior distribution over physical parameters given the PCA-compressed

observation:

$$p(\boldsymbol{\theta} \mid \mathbf{x}_{pca}) \propto p(\mathbf{x}_{pca} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}) \quad (37)$$

The likelihood $p(\mathbf{x}_{pca} \mid \boldsymbol{\theta})$ is the stochastic version of the PCA-compressed forward map, and $p(\boldsymbol{\theta})$ is the prior over physical parameters (uniform within the LHS bounds). Computing this posterior exactly is intractable; the VAE provides a scalable variational approximation.

5.5.2 Conditional VAE Formulation

We propose a **Conditional VAE (CVAE)** in which the latent prior is conditioned on the physical parameter labels. For Architecture Idea 2, L_d is retained as a target to allow its posterior variance to quantify its unobservability. The latent dimension is therefore 6. Define:

- **Encoder (Inference Network):** $q_\phi(\mathbf{z} \mid \mathbf{x}_{pca}) = \mathcal{N}(\boldsymbol{\mu}_\phi(\mathbf{x}_{pca}), \text{diag}(\boldsymbol{\sigma}_\phi^2(\mathbf{x}_{pca})))$, where $\boldsymbol{\mu}_\phi, \boldsymbol{\sigma}_\phi^2 \in \mathbb{R}^6$, output by a neural network with parameters ϕ .
- **Decoder (Generative Network):** $p_\psi(\mathbf{x}_{pca} \mid \mathbf{z}) = \mathcal{N}(\hat{\mathbf{z}}_{recon}, \sigma_{dec}^2 \mathbf{I})$, where $\hat{\mathbf{z}}_{recon} = \mathcal{D}_\psi(\mathbf{z}) \in \mathbb{R}^6$.
- **Conditional Prior:** $p(\mathbf{z} \mid \mathbf{Y}_{gt}) = \mathcal{N}(\mathbf{Y}_{gt}^{scaled}, \mathbf{I})$, a Gaussian centered at the normalized 6-dimensional ground-truth parameter vector.

The conditioning of the prior on $\mathbf{Y}_{gt}$ is the architectural key: it makes the KL divergence term itself a function of the ground-truth labels, aligning the statistical regularization force with the physics-supervision force.

Reparameterization Trick Sampling $\mathbf{z} \sim q_\phi(\mathbf{z} \mid \mathbf{x}_{pca})$ directly is non-differentiable. The reparameterization trick [4] resolves this by writing:

$$\mathbf{z} = \boldsymbol{\mu}_\phi(\mathbf{x}_{pca}) + \boldsymbol{\sigma}_\phi(\mathbf{x}_{pca}) \odot \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (38)$$

where $\odot$ denotes element-wise multiplication. This decomposition makes the sampling operation differentiable with respect to ϕ , allowing gradients to flow through $\mathbf{z}$ to the encoder.

5.5.3 Physics-Supervised CVAE Loss

The standard VAE trains by maximizing the Evidence Lower Bound (ELBO) on the log-likelihood:

$$\text{ELBO} = \mathbb{E}_{q_\phi(\mathbf{z} \mid \mathbf{X})} [\log p_\psi(\mathbf{X} \mid \mathbf{z})] - D_{KL}(q_\phi(\mathbf{z} \mid \mathbf{X}) \parallel p(\mathbf{z} \mid \mathbf{Y}_{gt})) \quad (39)$$

The first term is the reconstruction likelihood (equivalent to $-\mathcal{L}_{recon}$), and the second term is the KL divergence between the variational posterior and the conditional prior. For the Gaussian distributions defined above, the KL term has a closed form:

$$D_{KL}(\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2) \parallel \mathcal{N}(\mathbf{Y}_{gt}^{scaled}, \mathbf{I})) = \frac{1}{2} \sum_{k=1}^6 \left(\sigma_k^2 + (\mu_k - Y_{gt,k}^{scaled})^2 - 1 - \log \sigma_k^2 \right) \quad (40)$$

Note that Equation (40) *simultaneously* pulls $\boldsymbol{\mu}$ toward $\mathbf{Y}_{gt}^{scaled}$ (via the squared-error term $(\mu_k - Y_{gt,k}^{scaled})^2$) and regularizes the variance σ_k^2 to be neither too large nor collapsed to zero. This is the central advantage of the conditional prior: the physics information enters the KL term itself, aligning the two competing pressures (statistical regularization and physical accuracy) rather than placing them in opposition.

An additional explicit physics supervision term $\mathcal{L}_{phys} = \text{MSE}(\boldsymbol{\mu}, \mathbf{Y}_{gt}^{scaled})$ can optionally be included to provide a stronger gradient signal through the mean pathway:

$$\mathcal{L}_{total}^{VAE} = \mathcal{L}_{recon} + \beta D_{KL} + \lambda_1 \mathcal{L}_{phys} + \lambda_2 \mathcal{L}_{KK} \quad (41)$$

The hyperparameter $\beta > 0$ (following the β -VAE convention [4]) scales the KL term; $\beta < 1$ relaxes the constraint on the latent distribution and encourages better reconstruction at the cost of a less well-calibrated posterior.

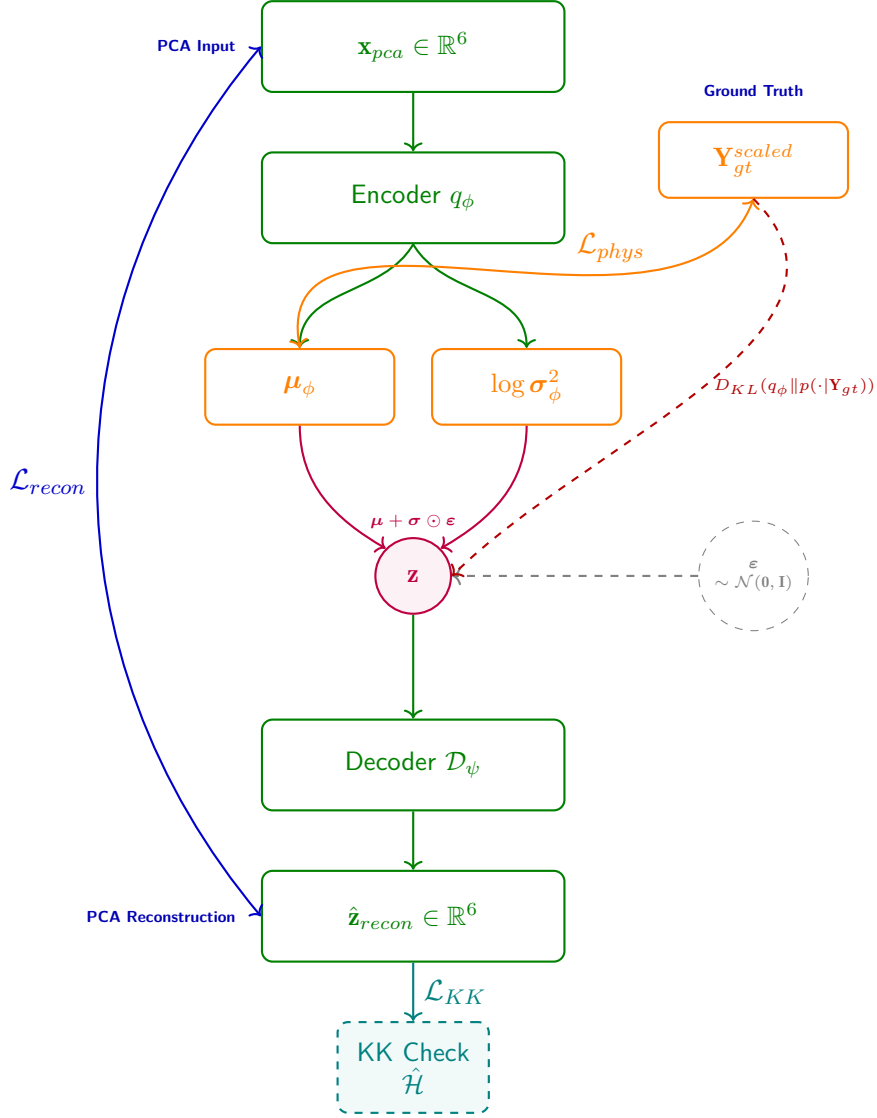


Figure 5: Architecture Idea 2: Physics-Supervised Conditional VAE. The encoder maps the 6D PCA input to a mean $\mu_\phi \in \mathbb{R}^6$ and log-variance $\log \sigma_\phi^2 \in \mathbb{R}^6$ over all 6 parameters including L_d ; the reparameterization trick (purple) allows differentiable sampling of $\mathbf{z}$. The conditional prior $p(\mathbf{z}|\mathbf{Y}_{gt}) = \mathcal{N}(\mathbf{Y}_{gt}^{scaled}, \mathbf{I})$ aligns the KL divergence force with the physics-supervision force $\mathcal{L}_{phys}$. The KK loss is applied after inverse-PCA projection of $\hat{\mathbf{z}}_{recon}$ to the full $\{S_{11}, S_{22}\}$ spectral domain. At inference time, σ_{L_d} quantifies the L_d unobservability.

5.5.4 Training Algorithm for the CVAE

Algorithm 2 Physics-Supervised Conditional VAE Training (Idea 2)

Require: Dataset $\mathcal{D} = \{(\mathbf{x}_{pca}^{(i)}, \mathbf{Y}^{(i)})\}_{i=1}^N$; fixed $\mathbf{V}_{:,1:6}$, $\boldsymbol{\mu}_{scaler}$; Epochs E ; Batch B ; LR α ; $\beta, \lambda_1, \lambda_2 \geq 0$

Ensure: Optimized encoder $(\boldsymbol{\mu}_\phi, \sigma_\phi^2)$ and decoder $\mathcal{D}_\psi$

- 1: Initialize ϕ, ψ using Xavier/Glorot initialization
- 2: **for** epoch $e = 1$ **to** E **do**
- 3: Shuffle $\mathcal{D}$
- 4: **for** each mini-batch $\{\mathbf{x}_b, \mathbf{Y}_b\} \subset \mathcal{D}$ **do**
- 5: $\boldsymbol{\mu}, \log \sigma^2 \leftarrow \text{Encoder}_\phi(\mathbf{x}_b)$
- 6: $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$; $\mathbf{z} \leftarrow \boldsymbol{\mu} + \exp(\frac{1}{2} \log \sigma^2) \odot \boldsymbol{\varepsilon}$
- 7: $\hat{\mathbf{z}}_{recon} \leftarrow \mathcal{D}_\psi(\mathbf{z})$ // 6D PCA reconstruction
- 8: // Inverse PCA for KK loss
- 9: $\hat{\mathbf{X}}_{1740} \leftarrow \mathbf{V}_{:,1:6} \hat{\mathbf{z}}_{recon} + \boldsymbol{\mu}_{scaler}$
- 10: Extract $(\hat{\mathbf{R}}_{11}, \hat{\mathbf{I}}_{11}, \hat{\mathbf{R}}_{22}, \hat{\mathbf{I}}_{22})$ from $\hat{\mathbf{X}}_{1740}$ per Eq. (1)
- 11: $\mathcal{L}_{recon} \leftarrow \frac{1}{B} \sum_j \|\mathbf{x}_b^{(j)} - \hat{\mathbf{z}}_{recon}^{(j)}\|^2$
- 12: $D_{KL} \leftarrow \frac{1}{2B} \sum_j \sum_{k=1}^6 \left(\sigma_{j,k}^2 + (\mu_{j,k} - Y_{gt,j,k}^{sc})^2 - 1 - \log \sigma_{j,k}^2 \right)$
- 13: $\mathcal{L}_{phys} \leftarrow \frac{1}{B} \sum_j \|\mathbf{Y}_b^{(j)} - \boldsymbol{\mu}^{(j)}\|^2$
- 14: $\mathcal{L}_{KK} \leftarrow \frac{1}{N_f} (\|\hat{\mathbf{R}}_{11} - \hat{\mathcal{H}}\{\hat{\mathbf{I}}_{11}\}\|^2 + \|\hat{\mathbf{I}}_{11} + \hat{\mathcal{H}}\{\hat{\mathbf{R}}_{11}\}\|^2$
- 15: $+ \|\hat{\mathbf{R}}_{22} - \hat{\mathcal{H}}\{\hat{\mathbf{I}}_{22}\}\|^2 + \|\hat{\mathbf{I}}_{22} + \hat{\mathcal{H}}\{\hat{\mathbf{R}}_{22}\}\|^2)$
- 16: $\mathcal{L}_{total}^{VAE} \leftarrow \mathcal{L}_{recon} + \beta D_{KL} + \lambda_1 \mathcal{L}_{phys} + \lambda_2 \mathcal{L}_{KK}$
- 17: $\phi \leftarrow \phi - \alpha \nabla_\phi \mathcal{L}_{total}^{VAE}$; $\psi \leftarrow \psi - \alpha \nabla_\psi \mathcal{L}_{total}^{VAE}$
- 18: **end for**
- 19: **end for**
- 20: **return** $\text{Encoder}_\phi, \mathcal{D}_\psi$

5.5.5 Inference and Uncertainty Quantification

At inference time, given a new measured PCA-compressed input $\mathbf{x}_{pca}^*$ (derived from a real VNA measurement via the fitted scaler and PCA transform), the CVAE produces:

- **Point estimate:** $\hat{\boldsymbol{\theta}} = \boldsymbol{\mu}_\phi(\mathbf{x}_{pca}^*)$, the MAP estimate of the 6 physical parameters. For the 5 observable parameters (R_b, C_b, R_d, C_d, R_a), this is a reliable extraction. For L_d , this will regress toward the prior mean with high uncertainty.
- **Uncertainty quantification:** $\sigma_\phi(\mathbf{x}_{pca}^*)$, the posterior standard deviation for each parameter. The L_d component is expected to exhibit a consistently large σ_{L_d} regardless of input, directly quantifying the null-space unobservability established in Section 5.1.5. High σ_k for observable parameters signals genuine degeneracy in that direction of parameter space.

- **Posterior sampling:** Drawing N_s samples $\mathbf{z}^{(s)} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2)$, passing each through $\mathcal{D}_\psi$, and inverse-PCA projecting yields a distribution over reconstructed $\{S_{11}, S_{22}\}$ spectra. This provides calibrated confidence intervals on the reconstructed spectra.

5.6 Model Interpretability and Physical Verification

Because both architectures explicitly supervise the bottleneck (or its mean, in the VAE case) to represent defined physical variables, the network serves as an ideal framework for mechanistic interpretability. The following interrogation methods test whether the network has internalized the underlying electromagnetic physics.

5.6.1 Latent Manifold Traversal with Smith Chart Geodesic Metric

The decoder $\mathcal{D}_\psi$ maps the 5-dimensional physical parameter space to the 6-dimensional PCA space, which via the fixed inverse-PCA projection produces the full $\{S_{11}, S_{22}\}$ spectral pair. If the decoder has truly learned the physics of RF reflection, sweeping a single parameter continuously through its range while holding the others fixed should trace a *smooth, physically predictable arc* on the Smith Chart for the S_{11} component of the reconstruction.

To quantify this, we introduce a rigorous geodesic deviation metric. The Smith Chart embeds the complex reflection coefficient $\Gamma = S_{11}$ in the unit disk $\mathbb{D} = \{\Gamma \in \mathbb{C} : |\Gamma| \leq 1\}$. The natural hyperbolic (Poincaré disk) metric on $\mathbb{D}$ is:

$$ds_{\mathbb{H}}^2 = \frac{4|d\Gamma|^2}{(1 - |\Gamma|^2)^2} \quad (42)$$

This induces a geodesic distance between two points $\Gamma_1, \Gamma_2 \in \mathbb{D}$:

$$d_{\mathbb{H}}(\Gamma_1, \Gamma_2) = 2 \tanh^{-1} \left(\frac{|\Gamma_1 - \Gamma_2|}{|1 - \bar{\Gamma}_1 \Gamma_2|} \right) \quad (43)$$

where $\bar{\Gamma}_1$ denotes the complex conjugate. This metric is invariant under Möbius transformations (the natural symmetries of the Smith Chart), making it the physically meaningful distance function on the reflection coefficient space.

The traversal and evaluation protocol proceeds as follows. For each parameter θ_m (e.g., R_d):

1. Generate $N_{trav} = 100$ uniformly spaced values $\theta_m^{(1)}, \dots, \theta_m^{(N_{trav})}$ spanning the full training range, while fixing all other parameters at nominal values.
2. For each $\theta_m^{(i)}$, obtain the **analytical** S_{11} curve directly from the circuit equations, and extract its value at the center frequency f_0 : $\Gamma_{truth}^{(i)} = S_{11}^{analytical}(f_0; \theta_m^{(i)})$.
3. For each $\theta_m^{(i)}$, query the decoder with the corresponding bottleneck vector, apply the inverse PCA, and extract the predicted S_{11} value at f_0 : $\hat{\Gamma}^{(i)} = S_{11}^{decoder}(f_0; \theta_m^{(i)})$.

4. Compute the cumulative geodesic deviation:

$$D_{geo}(\theta_m) = \frac{1}{N_{trav}} \sum_{i=1}^{N_{trav}} d_{\mathbb{H}}(\hat{\Gamma}^{(i)}, \Gamma_{truth}^{(i)}) \quad (44)$$

A well-trained physics surrogate yields $D_{geo} \rightarrow 0$ for all parameters. If a parameter exhibits high D_{geo} , it indicates that the decoder has not accurately internalized the circuit’s response to that parameter—a precise and quantitative failure diagnosis.

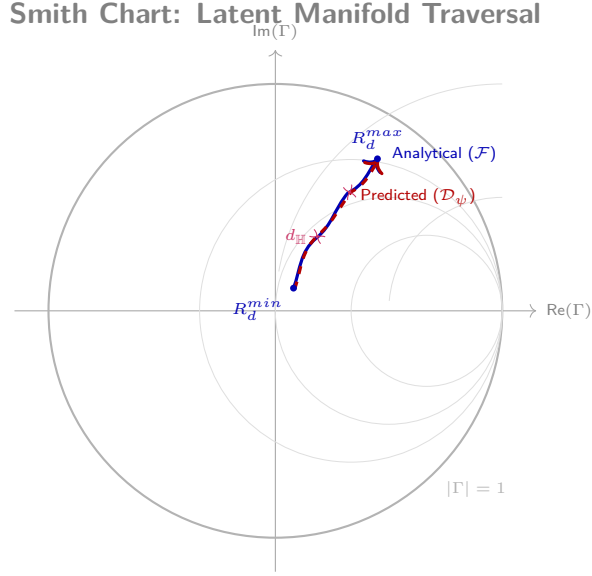


Figure 6: Smith Chart geodesic traversal for the R_d parameter. Sweeping R_d from R_d^{min} to R_d^{max} while holding all other parameters fixed traces a resistive locus. The geodesic deviation $d_{\mathbb{H}}$ (Equation (43)) measures the point-wise discrepancy between the analytical circuit trajectory and the decoder’s predicted trajectory in the Poincaré disk metric. A small cumulative D_{geo} (Equation (44)) is the quantitative proof that $\mathcal{D}_{\psi}$ has learned the electromagnetic equations rather than memorizing training data.

5.6.2 Frequency Attribution Mapping vs. Analytical Sensitivity

The encoder $\mathcal{E}_{\phi}$ extracts the 5 physical parameters from the 6-dimensional PCA input $\mathbf{x}_{pca}$. Since PCA components are linear mixtures of the full 1,740-dimensional spectral vector, the natural interpretability question is: *which frequency subbands drive the encoder’s predictions for each parameter?*

To answer this, Integrated Gradients [12] are computed with respect to the *original 1,740-dimensional input* $\mathbf{X}_{raw}$ rather than the 6D PCA space. This is achieved by propagating gradients through the full pipeline: $\mathbf{X}_{raw} \xrightarrow{\text{PCA}} \mathbf{x}_{pca} \xrightarrow{\mathcal{E}_{\phi}}$

$\hat{\theta}_m$. The saliency map $\mathbf{s}_m \in \mathbb{R}^{1740}$ for parameter θ_m is:

$$s_{m,j} = (X_{raw,j} - X_{raw,j}^{baseline}) \cdot \int_0^1 \frac{\partial \mathcal{E}_\phi(m) \circ \text{PCA}}{\partial X_{raw,j}} \Big|_{\mathbf{X}_{raw}^{baseline} + \alpha(\mathbf{X}_{raw} - \mathbf{X}_{raw}^{baseline})} d\alpha \quad (45)$$

where $\mathcal{E}_\phi(m)$ denotes the m -th encoder output. The saliency map is then averaged over the S_{11} and S_{22} channels and plotted as a function of frequency.

The neural saliency map $\mathbf{s}_m(f)$ is compared against the **analytical sensitivity**, derived directly from the circuit equations:

$$A_m(f) = \left| \frac{\partial Z_{in}(f; \boldsymbol{\theta})}{\partial \theta_m} \right| \quad (46)$$

A strong correlation between the neural saliency profile and the analytical sensitivity profile (as a function of frequency) is direct, quantitative proof that the encoder’s feature extraction strategy is physically grounded. The expected structure, consistent with the PC–parameter correlations of Section 5.1, is: high saliency at low-to-mid GHz for R_d and R_b ; moderate-frequency dominance for C_d and C_b ; flat frequency saliency for R_a .

5.6.3 Out-of-Distribution Extrapolation

A standard MLP is famously poor at extrapolating outside its training distribution. Because the proposed architecture is strongly physics-informed via the bottleneck supervision and the KK prior, the hypothesis is that the decoder has approximated the actual circuit equations and can therefore generalize beyond the training range.

To test this, the decoder is fed isolated latent vectors with parameter values outside the LHS training bounds. For example, if trained on $R_d \in [2 \text{ k}\Omega, 25 \text{ k}\Omega]$, the decoder is queried at $R_d = 35 \text{ k}\Omega$ while all other parameters remain at nominal in-distribution values. The decoder’s 6-dimensional PCA output is projected back to the full $\{S_{11}, S_{22}\}$ spectrum via the inverse-PCA transform, and the result is compared against a direct ADS simulation at the same OOD parameter values. The geodesic deviation metric D_{geo} defined in Section 5.6.1 is directly applicable as a quantitative OOD performance score: a model that has internalized the electromagnetic equations will maintain low D_{geo} outside the training bounds, whereas a statistical interpolator will exhibit rapidly diverging D_{geo} as the query point moves away from the training distribution.

5.7 Planned Ablation Study Protocol

To rigorously attribute performance improvements to specific architectural decisions, the following ablation study protocol is proposed. All models are trained and evaluated on identical train/test splits of the 20,000-sample LHS dataset (80/20 split), with a held-out OOD test set of 500 samples generated with all six variable parameter values uniformly distributed 20% outside the LHS training bounds.

5.7.1 Baseline and Comparison Models

1. **Levenberg–Marquardt (LM) Circuit Fitting:** Classical gradient-based optimization of the 6-parameter circuit model $(R_b, C_b, R_d, L_d, C_d, R_a)$ — fixed extrinsic elements held at reference values) directly against each test $\{S_{11}, S_{22}\}$ pair. This is the standard industrial method and serves as the physical-accuracy ceiling and speed-floor baseline. Note that LM operates on the raw 1,740-dimensional spectral data, not on the PCA-compressed representation.
2. **Direct MLP Regression (PCA input):** A fully-connected MLP trained with supervised regression directly from $\mathbf{x}_{pca} \in \mathbb{R}^6$ to $\mathbf{Y} \in \mathbb{R}^5$ ($\mathcal{L}_{phys}$ only, no decoder, no KK prior). Tests whether the autoencoder structure adds value over simple regression on PCA features.
3. **Unsupervised Autoencoder (No Physics):** Standard autoencoder with $\mathcal{L}_{recon}$ only and no bottleneck supervision. The bottleneck outputs are mapped to $\boldsymbol{\theta}$ via a post-hoc linear probe. Tests whether reconstruction alone induces a physically meaningful latent structure in the 6D PCA space.
4. **PIAE Dual-Loss ($\lambda_2 = 0$):** Architecture Idea 1 with only $\mathcal{L}_{recon} + \lambda_1 \mathcal{L}_{phys}$. Isolates the incremental contribution of the Kramers–Kronig causality term.
5. **PIAE Full Triple-Loss (Idea 1):** Complete deterministic architecture with $\mathcal{L}_{recon} + \lambda_1 \mathcal{L}_{phys} + \lambda_2 \mathcal{L}_{KK}$, using the λ selection strategy of Section 5.4.2.
6. **Physics-Supervised CVAE (Idea 2):** Complete variational architecture trained with $\mathcal{L}_{total}^{VAE}$ (Equation (41)), retaining L_d as a target to quantify its posterior uncertainty.

5.7.2 Evaluation Metrics

For each model, the following metrics are reported on the in-distribution test set and the OOD test set separately:

- **Per-parameter MAPE:** $\text{MAPE}_k = \frac{100\%}{N_{test}} \sum_i \left| \frac{\hat{\theta}_k^{(i)} - \theta_k^{(i)}}{\theta_k^{(i)}} \right|$ for each $k \in \{R_b, C_b, R_d, C_d, R_a\}$. Relative percentage error accounts for the order-of-magnitude differences between parameters. L_d is excluded from MAPE reporting for Idea 1 (fixed) and reported as posterior variance for Idea 2 (CVAE).
- **Geodesic Deviation D_{geo} :** The Smith Chart traversal metric (Equation (44)) reported for each of the 5 extractable parameters.
- **KK Consistency Residual:** $\mathcal{L}_{KK}$ evaluated on the reconstructed spectra of the test set after inverse-PCA projection. Lower is better and indicates more physically causal decoder outputs.

- **Posterior Calibration (Idea 2 only):** For the CVAE, the ratio $\sigma_{L_d}/\bar{\sigma}_{others}$ quantifies how much more uncertain the model is about L_d relative to the other parameters. A large ratio is the expected signature of the null-space finding from Section 5.1.5.
- **Inference Throughput:** Wall-clock time per parameter extraction, demonstrating the speed advantage of the neural surrogate over LM fitting.

5.7.3 Expected Ablation Findings

The following outcomes are hypothesized based on the architectural analysis:

- The unsupervised autoencoder (Model 3) is expected to achieve low $\mathcal{L}_{recon}$ but high per-parameter MAPE, since a 6D PCA autoencoder has no incentive to organize its latent dimensions along physical parameter directions without explicit supervision.
- Adding $\mathcal{L}_{phys}$ (Models 4 and 5 vs. Model 3) is expected to dramatically reduce MAPE for the four well-conditioned parameters (R_a , R_d , R_b , C_b), confirming bottleneck supervision as the primary driver of physical accuracy.
- Adding $\mathcal{L}_{KK}$ (Model 5 vs. Model 4) is expected to improve OOD D_{geo} scores and reduce the KK residual on the test set, since the causality prior discourages the decoder from overfitting to training-distribution spectral patterns. The improvement on in-distribution MAPE may be modest, but OOD generalization should show a clear benefit.
- The CVAE (Model 6) is expected to match Model 5 on point-estimate MAPE for the 5 observable parameters, while providing a quantitatively large σ_{L_d} that correctly reports the unobservability of L_d .

The planned ablation table (to be populated upon experiment completion) is structured as follows:

Table 1: Planned ablation study results table. Values to be populated upon completion. †: CVAE reports posterior σ_{L_d} in place of MAPE for L_d .

Model	MAPE $_{R_d}$	MAPE $_{R_b}$	MAPE $_{C_d}$	MAPE $_{R_a}$	D_{geo} (avg.)	KK Resid.	OOD MAPE
LM Fitting	—	—	—	—	—	—	—
Direct MLP (PCA)	—	—	—	—	—	—	—
Unsup. AE	—	—	—	—	—	—	—
PIAE (dual-loss)	—	—	—	—	—	—	—
PIAE (triple, Idea 1)	—	—	—	—	—	—	—
CVAE† (Idea 2)	—	—	—	—	—	—	—

6 Conclusion and Future Work

This document outlines a rigorous, physics-first methodology for the extraction of parasitic parameters in GaN-based III-V semiconductor diodes using deep

learning. The methodology is structured as a sequential pipeline in which each stage corrects a specific physical or mathematical misalignment that would otherwise corrupt the machine learning model.

The circuit is operated as a 2-port network with four extrinsic elements ($C_{pad}, L_{pad}, R_L, C_{ct}$) held at reference values, and six variable intrinsic elements ($R_b, C_b, R_d, L_d, C_d, R_a$) serving as extraction targets. An empirical correlation analysis on the simulated $\{S_{11}, S_{21}, S_{22}\}$ dataset establishes that S_{21} and S_{22} are linearly identical ($|r|_{Re} = |r|_{Im} = 1.000$), and S_{21} is dropped. The retained channels S_{11} and S_{22} are concatenated in Cartesian real-imaginary form to produce a phase-wrap-free 1,740-dimensional raw input.

A Joint Complex PCA — mathematically equivalent to Widely Linear PCA for the non-circular signals observed here ($|r|_{Re} = 0.92$ vs. $|r|_{Im} = 0.58$ between channels) — compresses this 1,740-dimensional vector to just 6 principal components capturing $\approx 100\%$ of the total variance. This compression is lossless because the 6-parameter SSEC forward map generates a 6-dimensional data manifold in the ambient space, confirmed by the near-diagonal PC-parameter correlation structure. A critical empirical finding from this analysis is the **unobservability of L_d** : its maximum correlation with any principal component is $|r| = 0.10$, establishing that L_d lies in the effective null space of the measurement operator across the target frequency range. This is explained analytically by the negligibility of ωL_d relative to R_d throughout the 40 MHz–43.5 GHz band.

The six-dimensional curse of dimensionality (which would require $20^6 = 6.4 \times 10^7$ grid search simulations) is resolved through Latin Hypercube Sampling, generating a dense, space-filling 20,000-sample dataset. The parameter extraction problem is formally characterized as an ill-posed inverse problem via Jacobian analysis: the $\mathcal{L}_{phys}$ bottleneck supervision term is identified as a Tikhonov regularizer that stabilizes the inverse map, not merely a convenience for label assignment. The Kramers–Kronig consistency loss $\mathcal{L}_{KK}$ enforces electromagnetic causality on the decoder’s output through a differentiable inverse-PCA bridge, operating jointly on the S_{11} and S_{22} channels. The λ_1, λ_2 selection problem is recast as a Lagrangian-constrained optimization with non-negativity as a necessary KKT condition, clarifying why hand-tuned negative weights are physically nonsensical and motivating three principled selection strategies.

The conditional VAE (Architecture Idea 2) extends the deterministic framework to full Bayesian posterior inference: its inflated σ_{L_d} output is the probabilistic manifestation of the PCA null-space finding, correctly reporting unobservability rather than hallucinating a spurious point estimate. The mechanistic interpretability framework — Smith Chart geodesic deviation D_{geo} and Integrated Gradients saliency benchmarked against analytical circuit sensitivities — provides quantitative tests distinguishing a genuine physics surrogate from a statistical interpolator.

Immediate next steps are: implementing and debugging the ADS automation pipeline; generating the full 20,000-sample LHS dataset for the 6-parameter circuit; implementing both architectures in PyTorch with the FFT-based dual-channel KK loss; executing the λ grid search and Kendall weighting experiments; and populating the ablation table of Section 5.7.

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